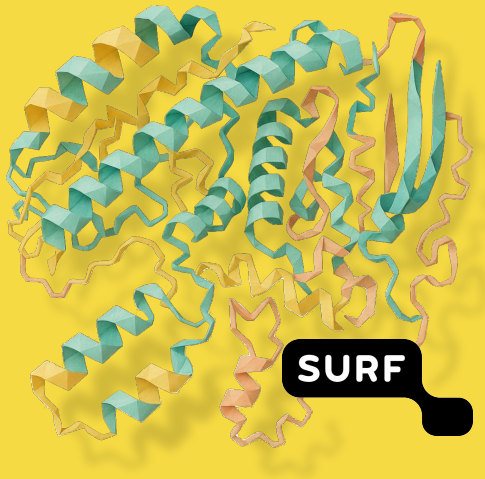
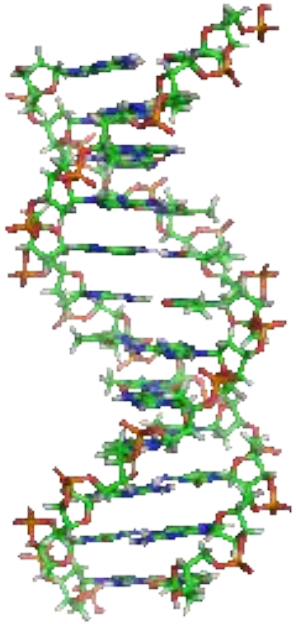
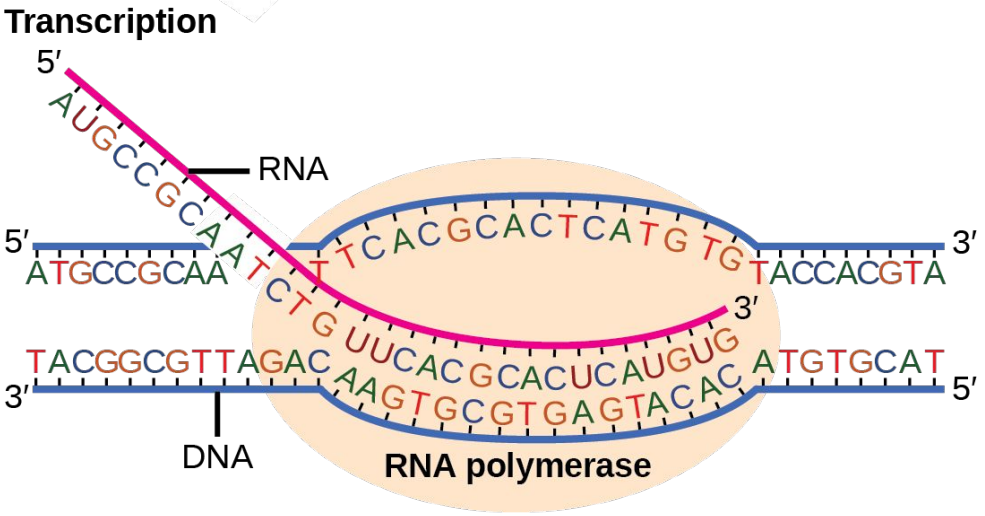
Protein Folding & Design

Bryan Cardenas
High Performance Machine Learning





ACTG

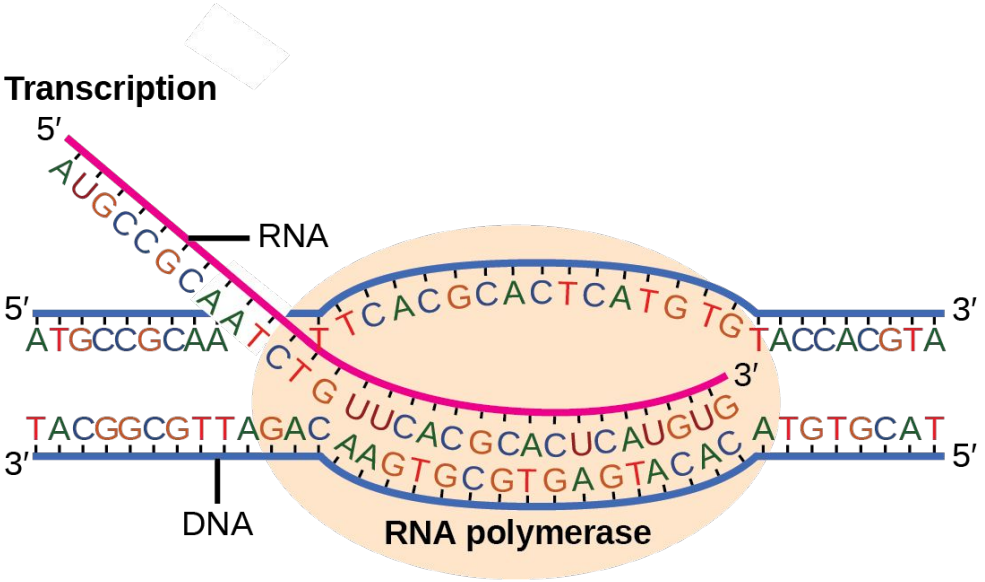

Transcription
5′
AUGCCGCAA
RNA
5′
ATGCCGCAA
TTCACGCACTCATGTGTACCACGTA
3′
3′
TACGGCGTTAGAC
AAGTGCGTGAGTACAC
ATGTGCAT
5′
3′
DNA
RNA polymerase

1. **Transcription**
   Open, read, copy (mRNA)

2. **Translation**
   read, fetch (tRNA), create protein

   Generate protein until stop codon

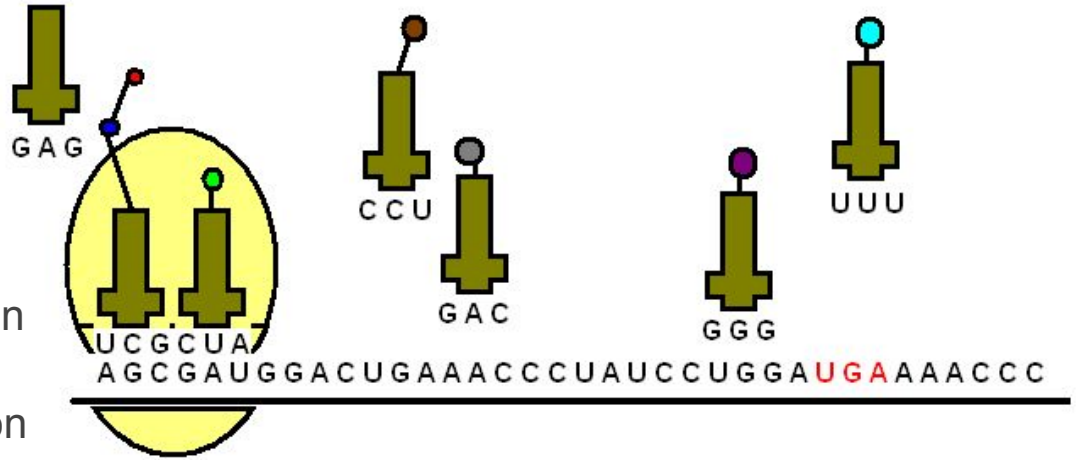
ACTG



1. **Transcription**
   Open, read, copy (mRNA)

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   Generate protein until stop codon



ACTG

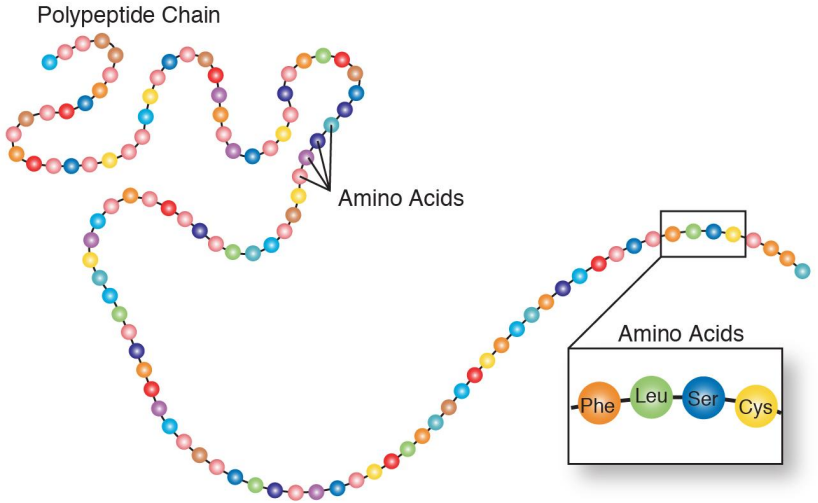
1. Transcription

Open, read, copy (mRNA)

2. Translation

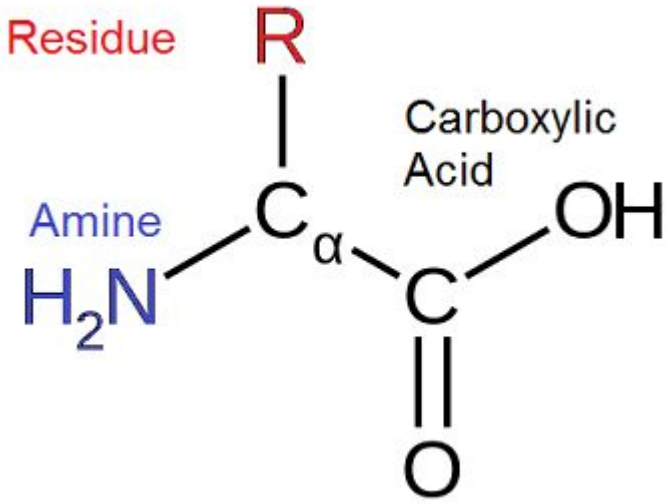
read, fetch (tRNA), create protein

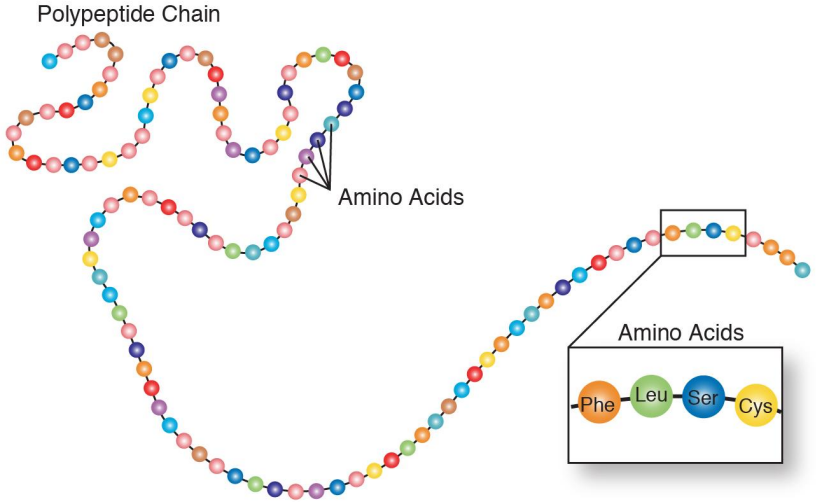
Generate protein until stop codon



Amino Acids

Ala: Alanine	Gln: Glutamine	Leu: Leucine	Ser: Serine
Arg: Arginine	Glu: Glutamic acid	Lys: Lysine	Thr: Threonine
Asn: Asparagine	Gly: Glycine	Met: Methionine	Trp: Tryptophane
Asp:Aspartic acid	His: Histidine	Phe: Phenylalanine	Tyr: Tyrosisne
Cys:Cysteine	Ile: Isoleucine	Pro: Proline	Val: Valine

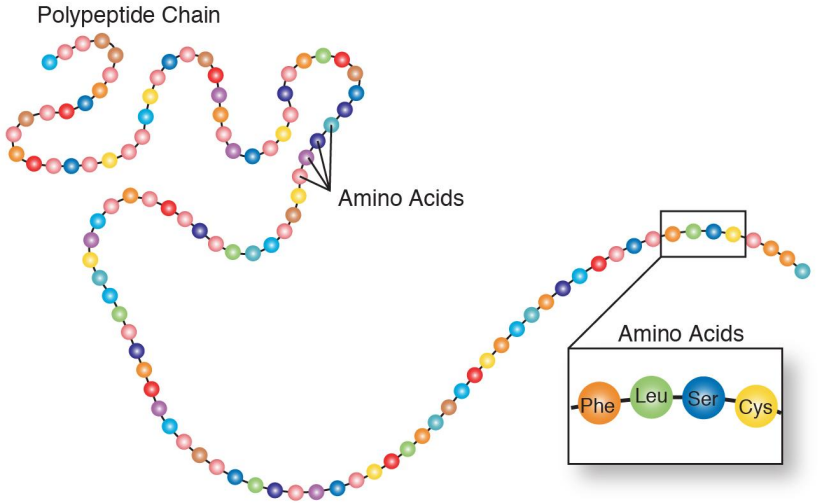




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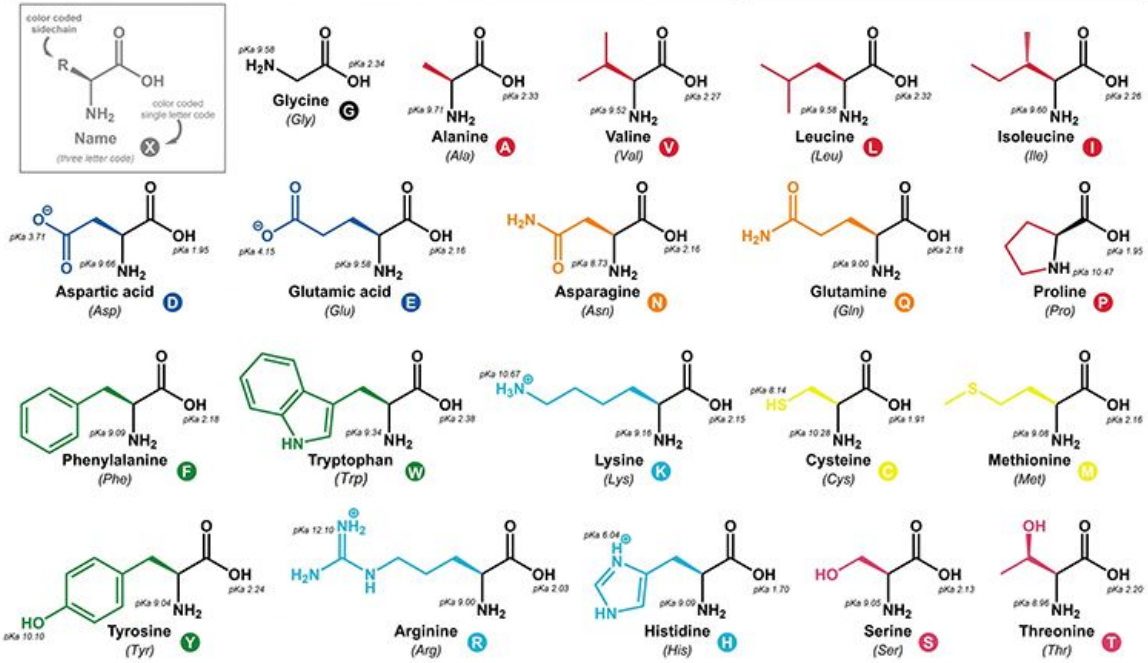
Amino Acids

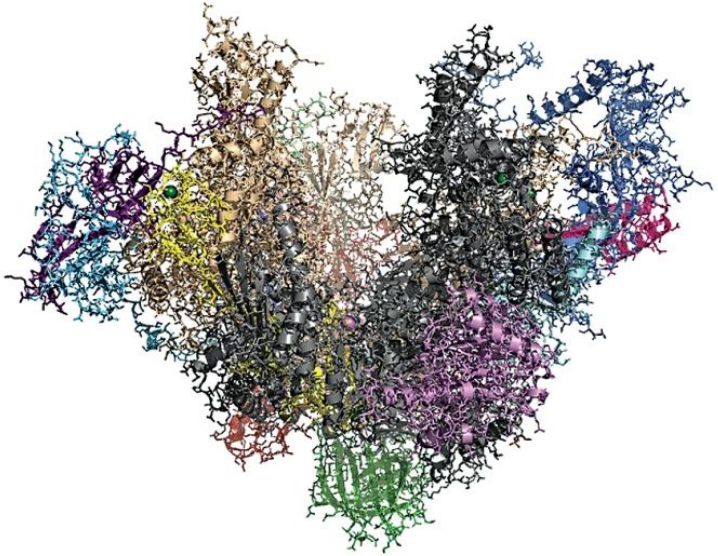
Ala: Alanine	Gln: Glutamine	Leu: Leucine	Ser: Serine
Arg: Arginine	Glu: Glutamic acid	Lys: Lysine	Thr: Threonine
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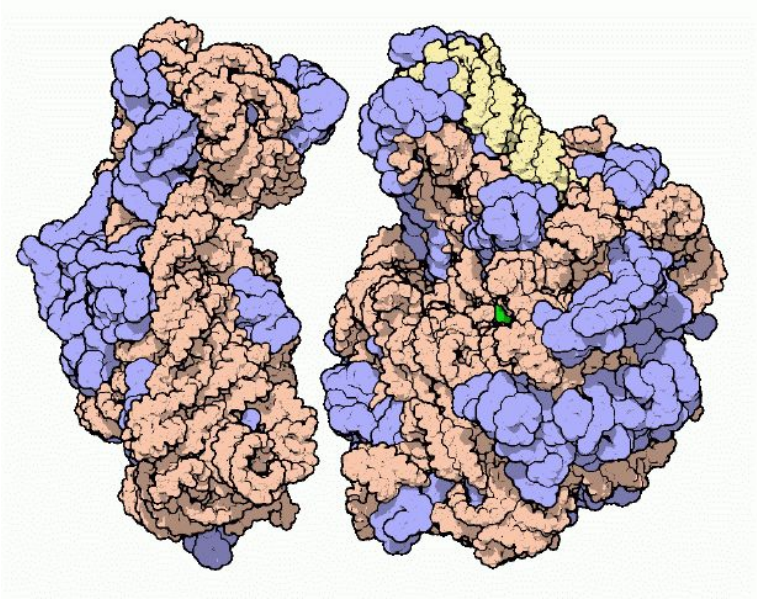
THE 20 COMMON AMINO ACIDS





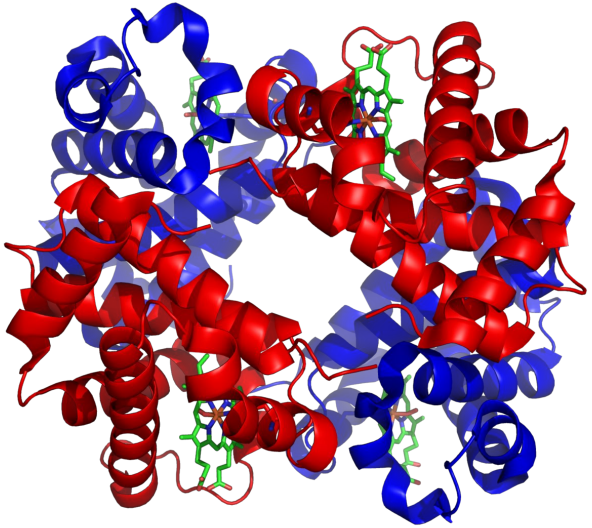


RNA Polymerase

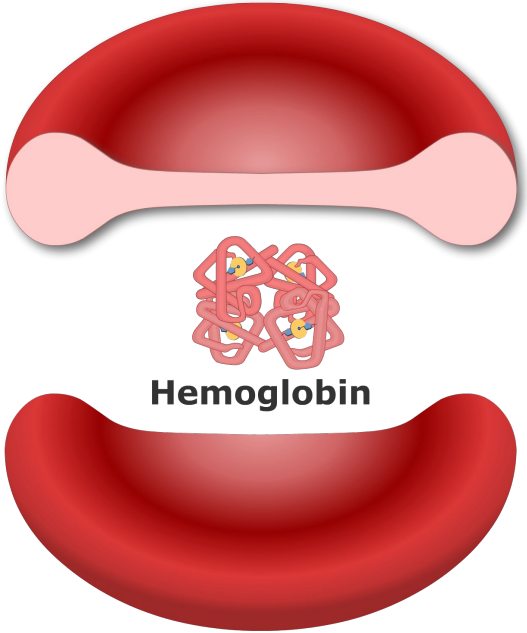


Ribosome

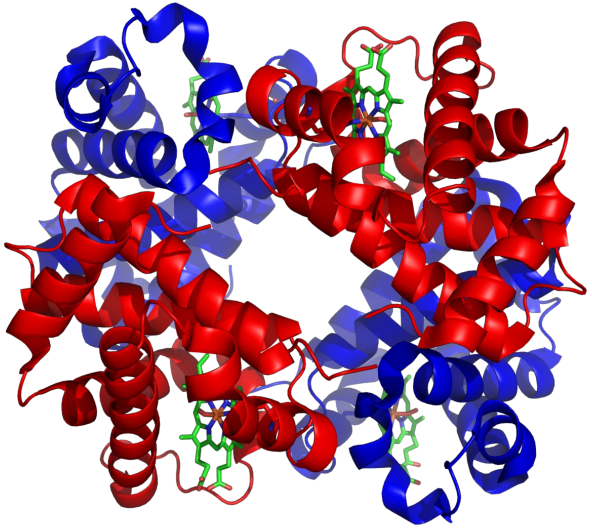
Hemoglobin



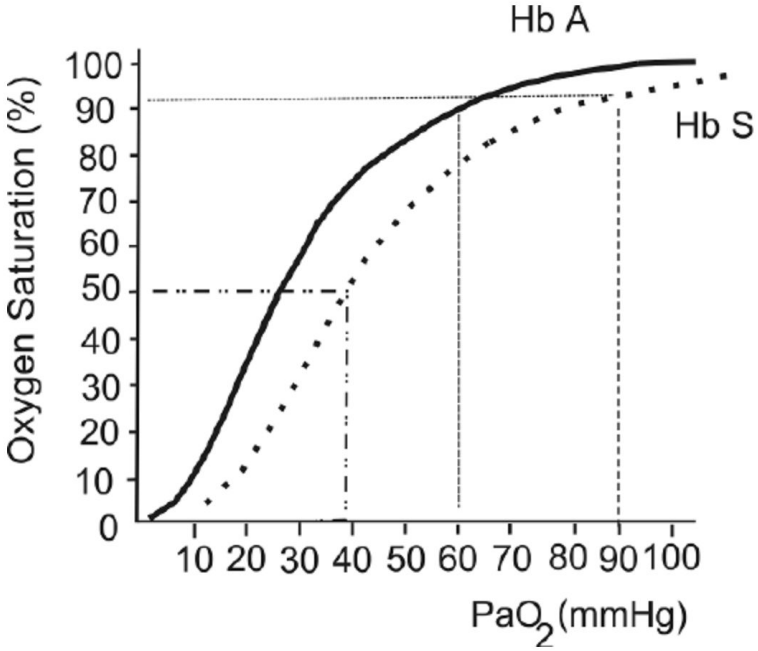
Heterotetramer



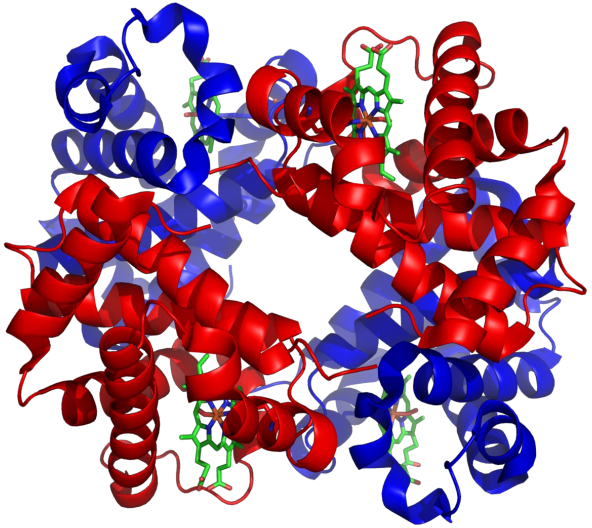
Hemoglobin



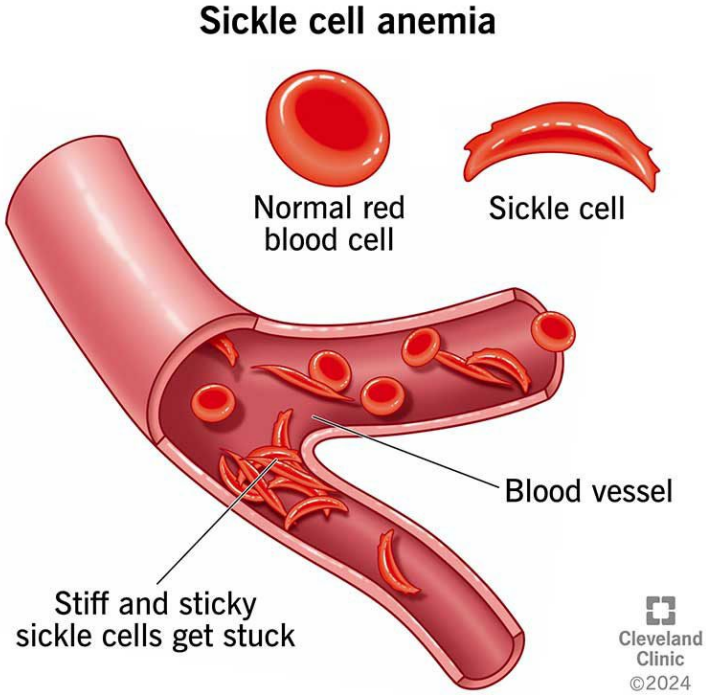
Heterotetramer



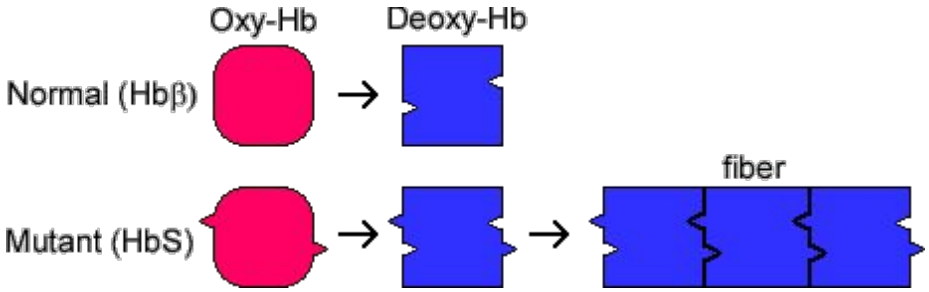
Hemoglobin

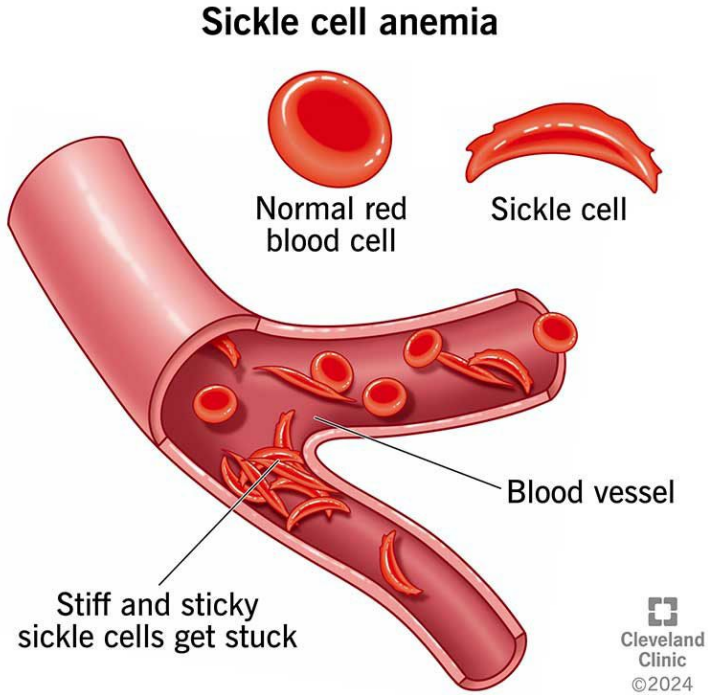


Heterotetramer

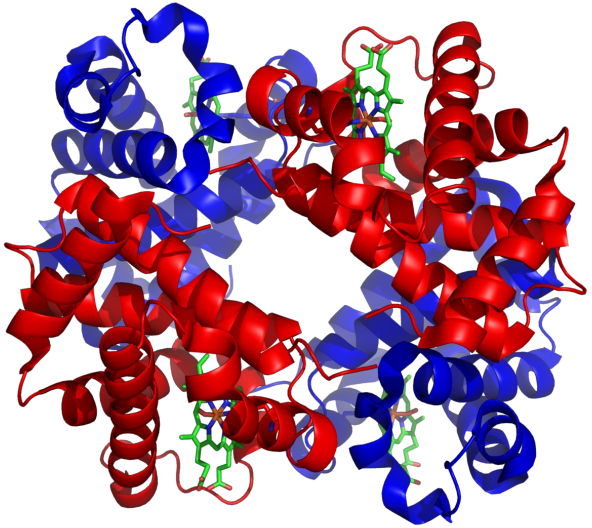


Hemoglobin

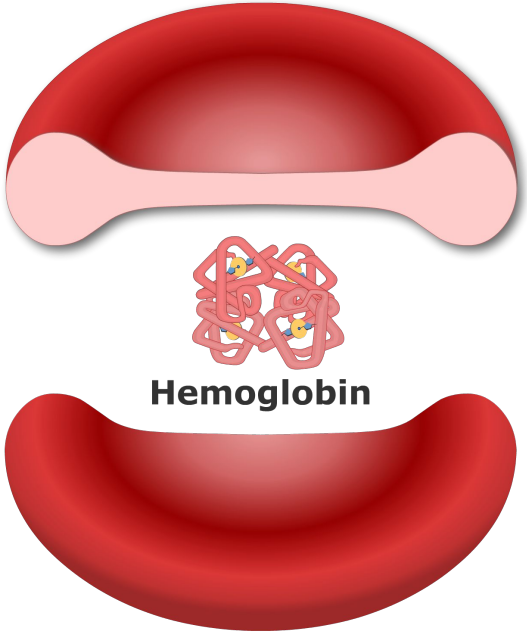

Oxy-Hb
Deoxy-Hb
Normal (Hbβ)
fiber
Mutant (HbS)


Sickle cell anemia
Normal red blood cell
Sickle cell
Blood vessel
Stiff and sticky sickle cells get stuck
Cleveland Clinic
©2024

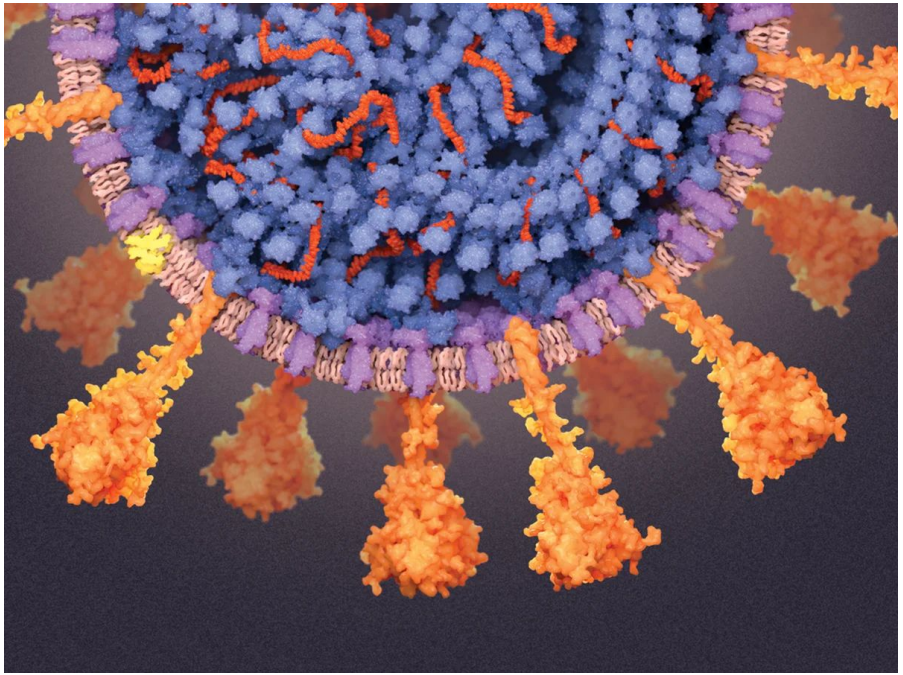
Hemoglobin



Heterotetramer

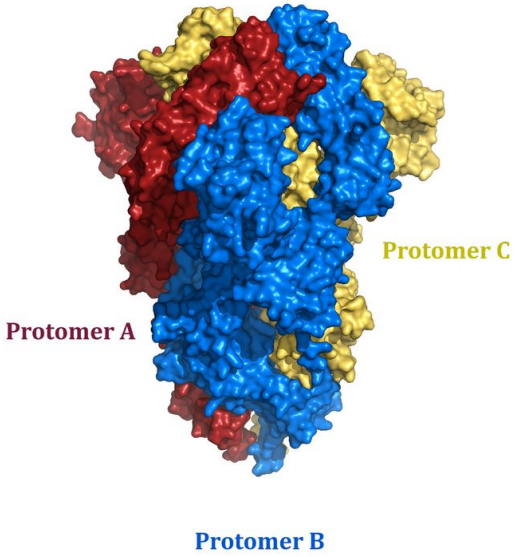


SARS-COV-2 Spike glycoprotein homotrimer

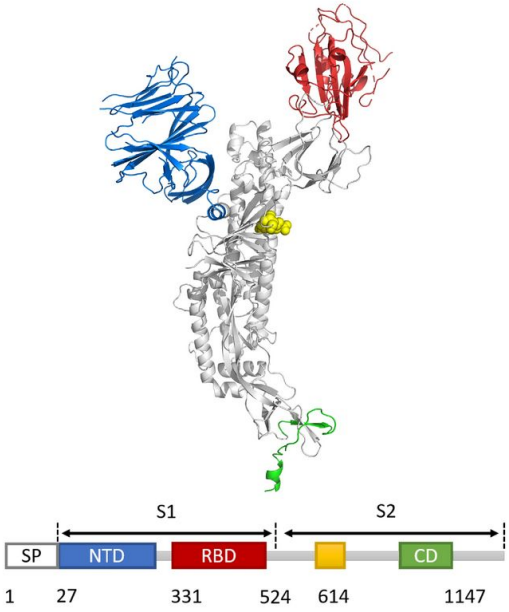


SARS-COV-2 Spike glycoprotein homotrimer

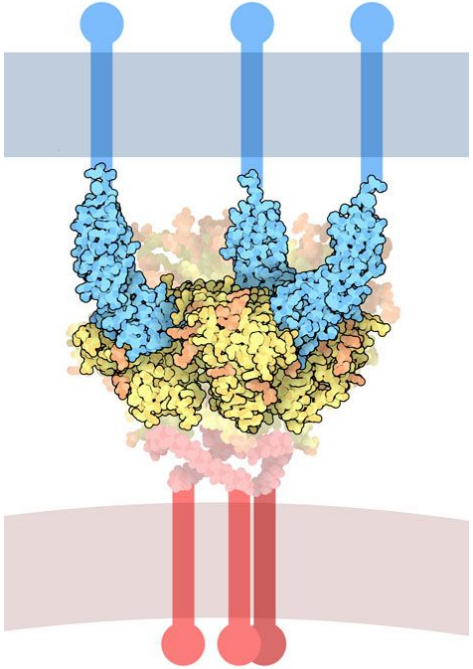
(a) Spike Trimer Protein



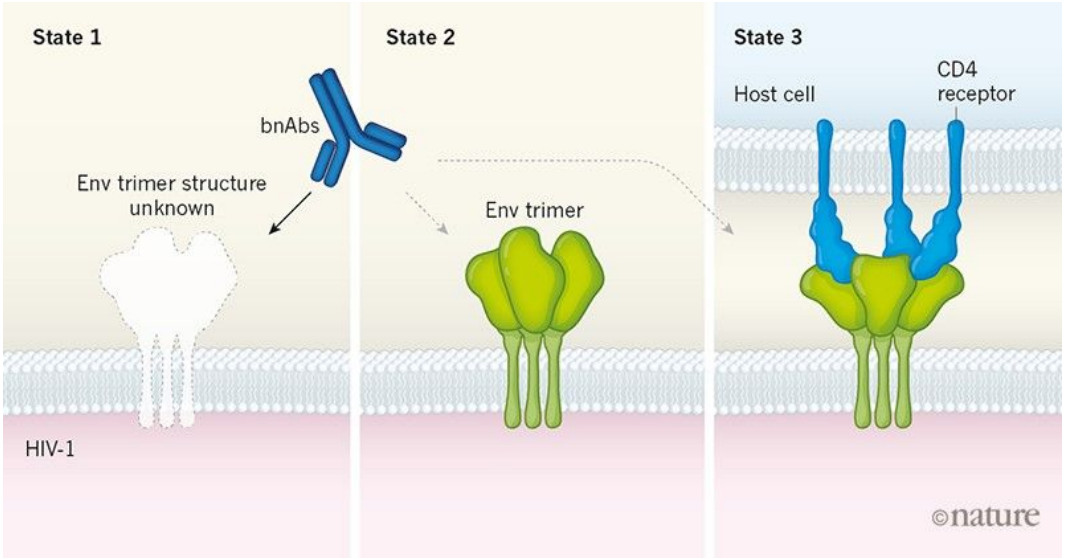
(b) Single Spike Protomer



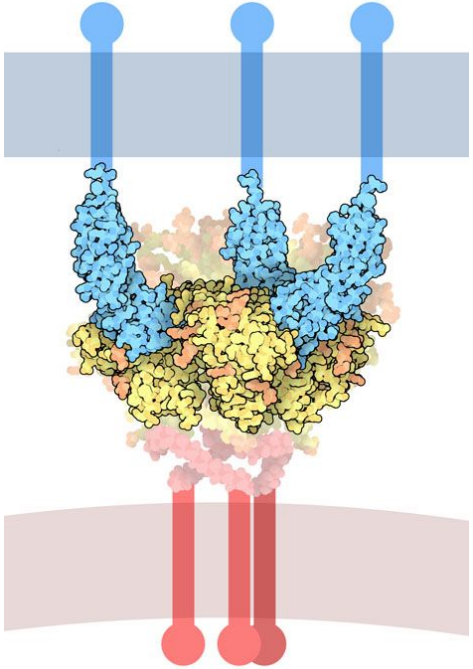
Hiv-1 Env Glycoprotein trimer



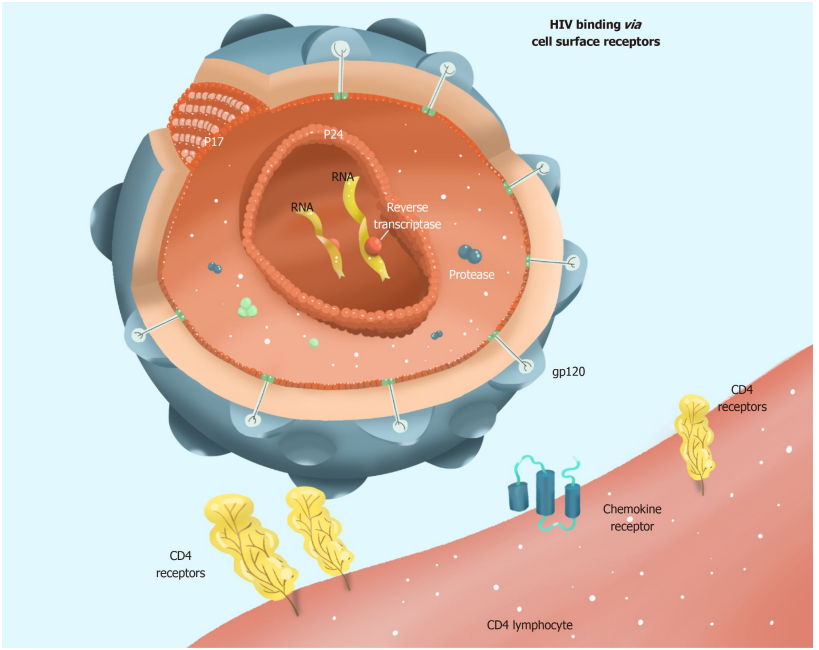
Trimer



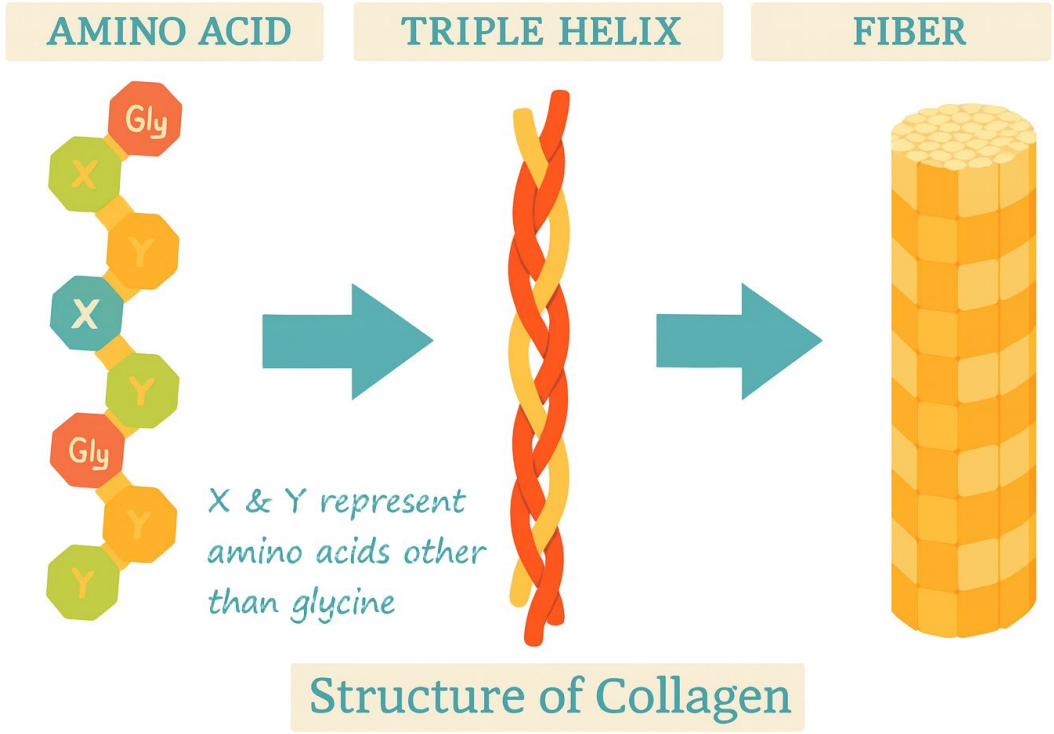
Hiv-1 Env Glycoprotein trimer



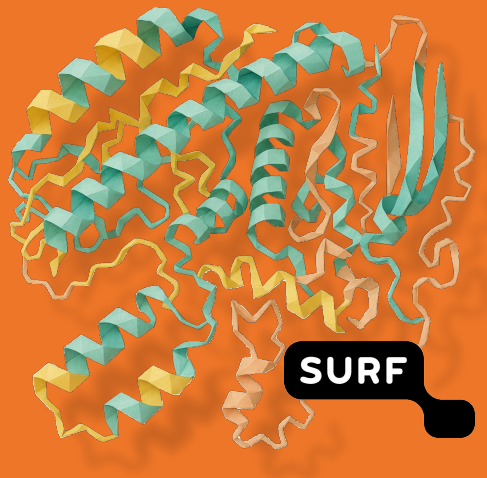
Trimer



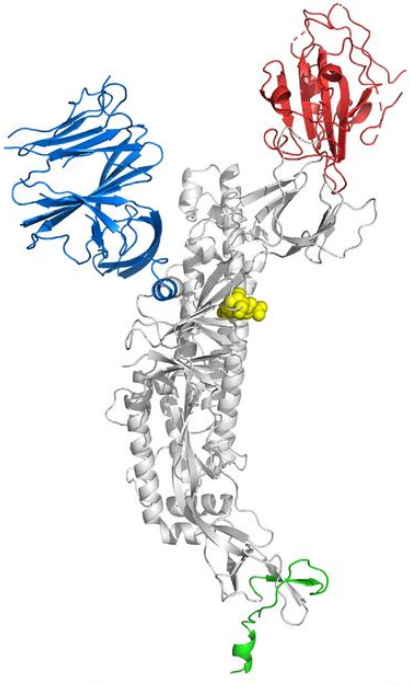
Structure of Collagen



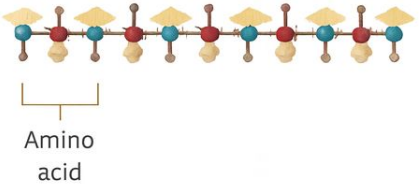
Evolution of Structure



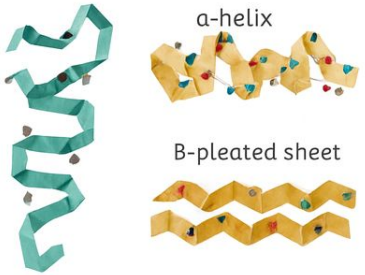
Hierarchy of Structure



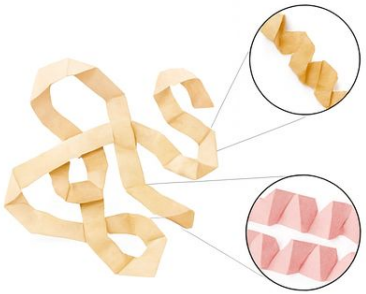
Primary structure



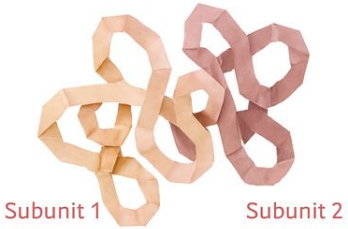
Sec·ondary structure



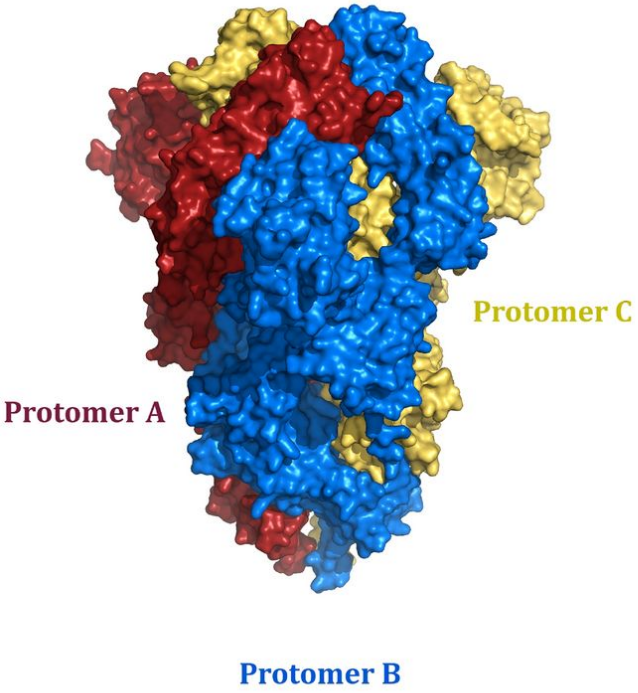
Tertiary structure



Quaternary structure



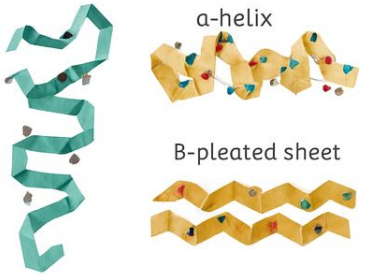
Hierarchy of Structure



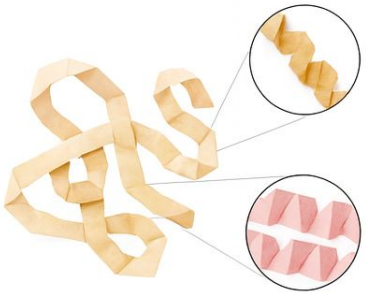
Primary structure



Sec·ondary structure



Tertiary structure



Quaternary structure



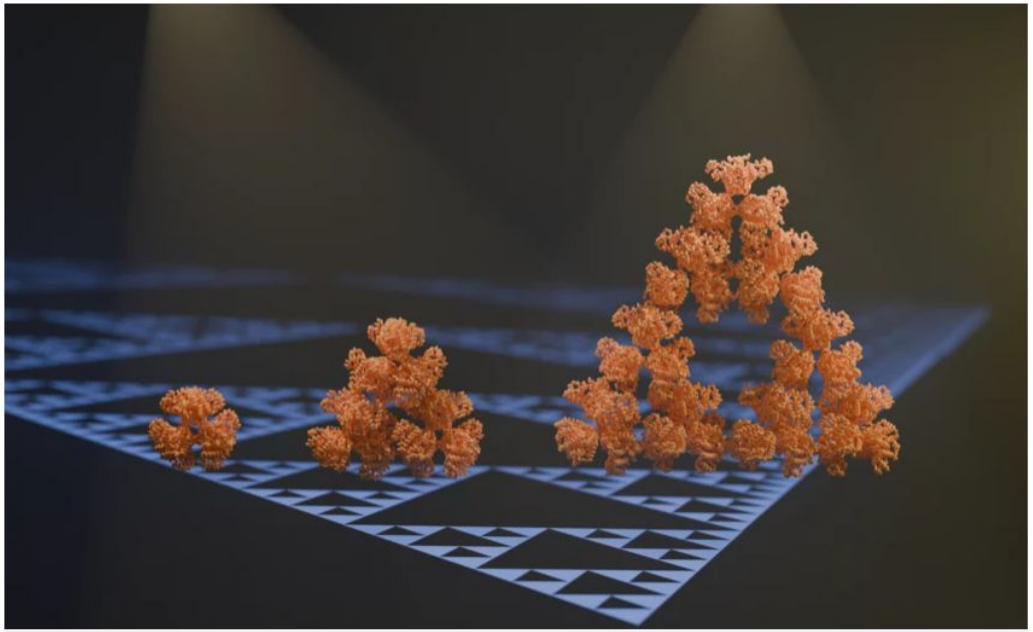
Discovery of the first fractal molecule in nature

Scientists found for the first time a natural protein that follows a mathematical pattern of self-similarity

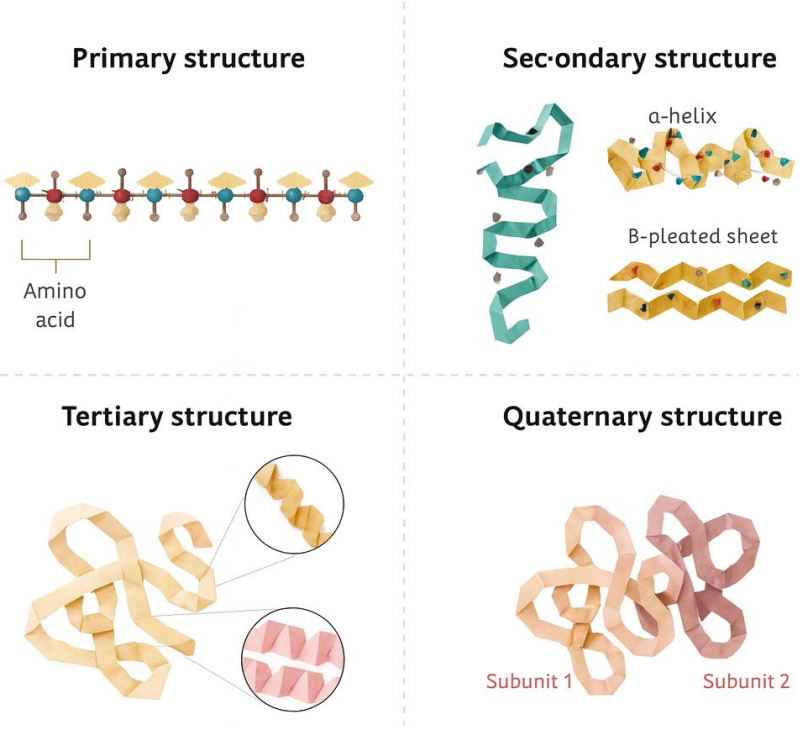
APRIL 10, 2024

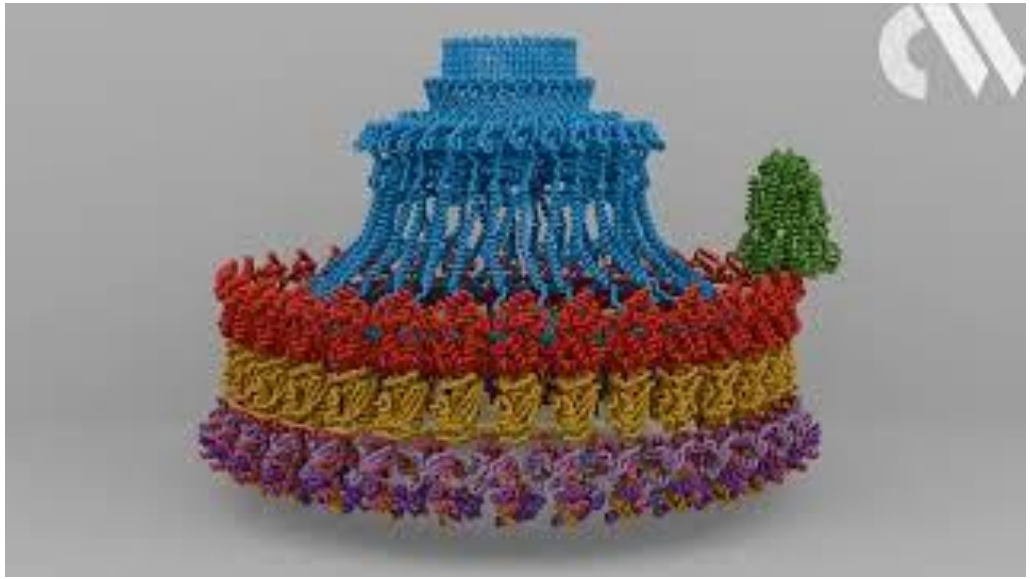
Proteins | Structural Biology

An international team of researchers led by groups from the Max Planck Institute in Marburg and the Philipps University in Marburg has stumbled upon the first regular molecular fractal in nature. They discovered a microbial enzyme - citrate synthase from a cyanobacterium - that spontaneously assembles into a pattern known as the Sierpinski triangle. Electron microscopy and evolutionary biochemistry studies indicate that this fractal may represent an evolutionary accident.



Citrate synthase spontaneously assembles into a fractal pattern known as the Sierpinski triangle.

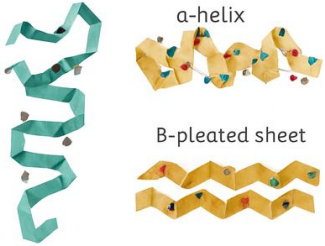




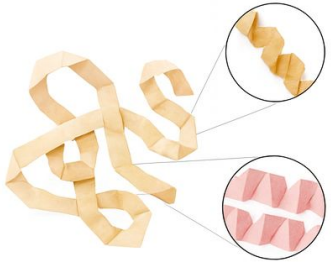
Primary structure



Sec·ondary structure



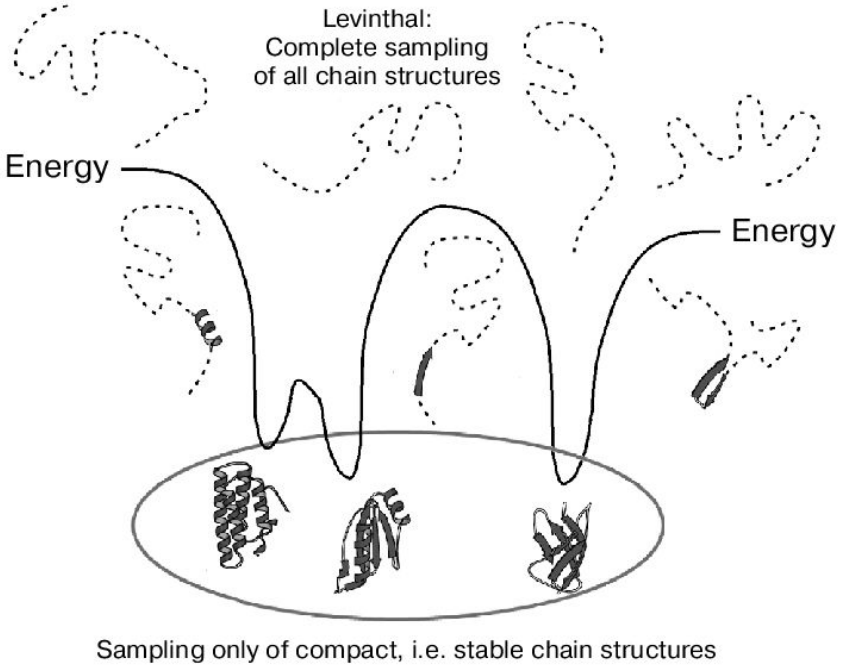
Tertiary structure



Quaternary structure

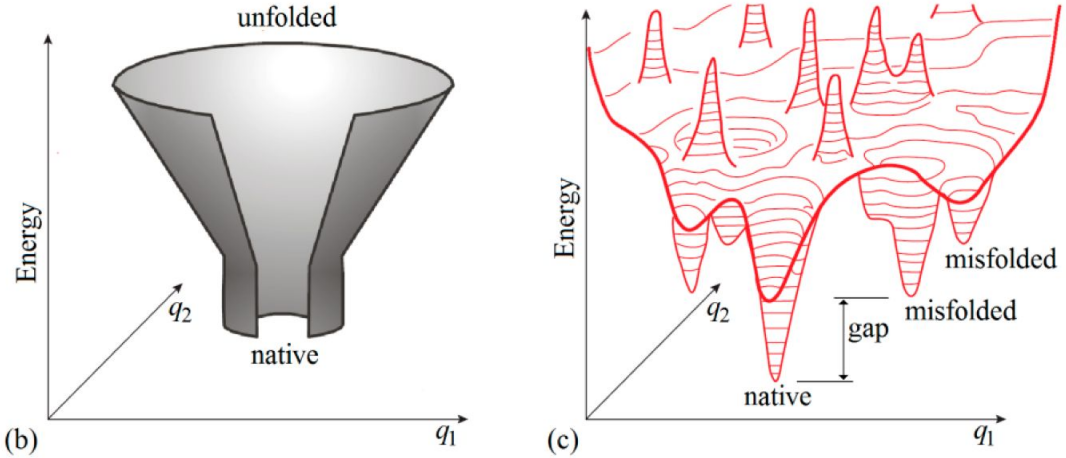


Levinthal's paradox



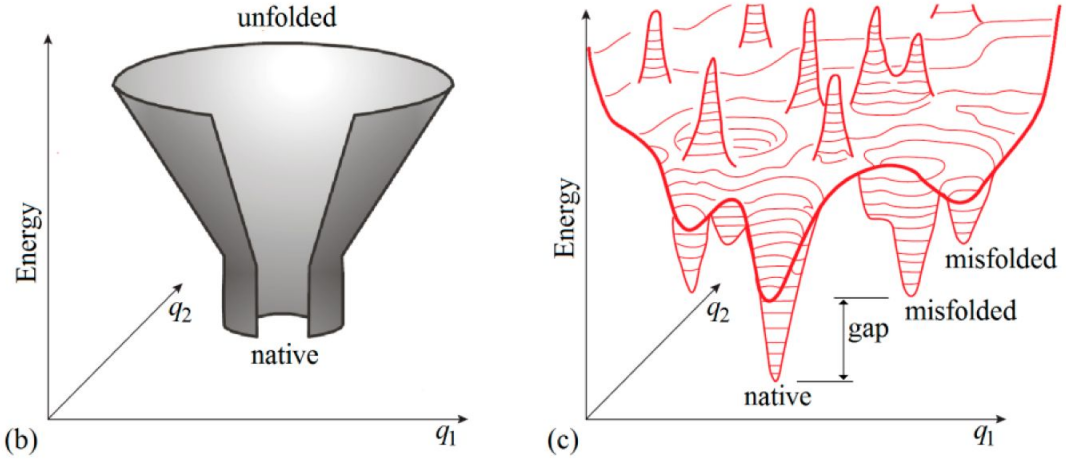
1. Proteins can fold in a astronomical number of different ways

Levinthal's paradox



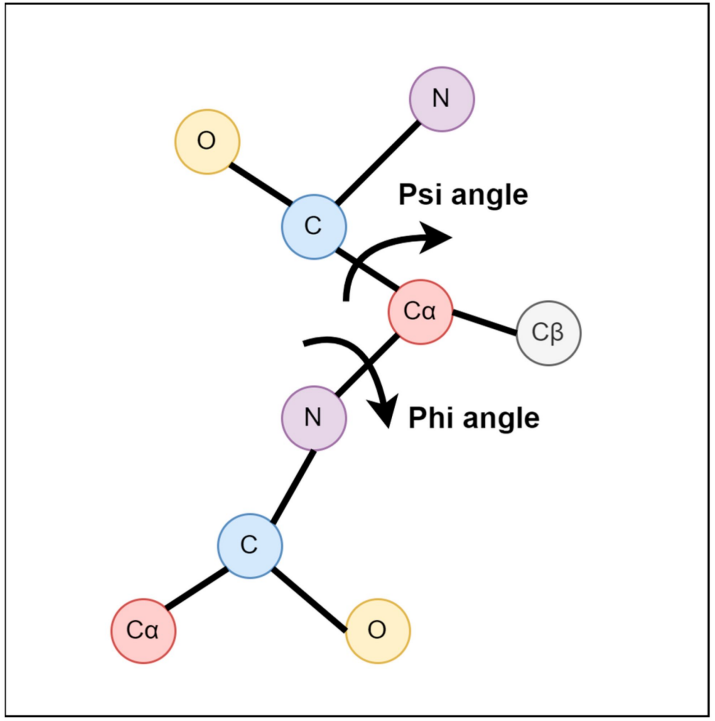
1. Proteins can fold in a astronomical number of different ways
2. Protein always finds a stable minimum of the folding landscape

Levinthal's paradox

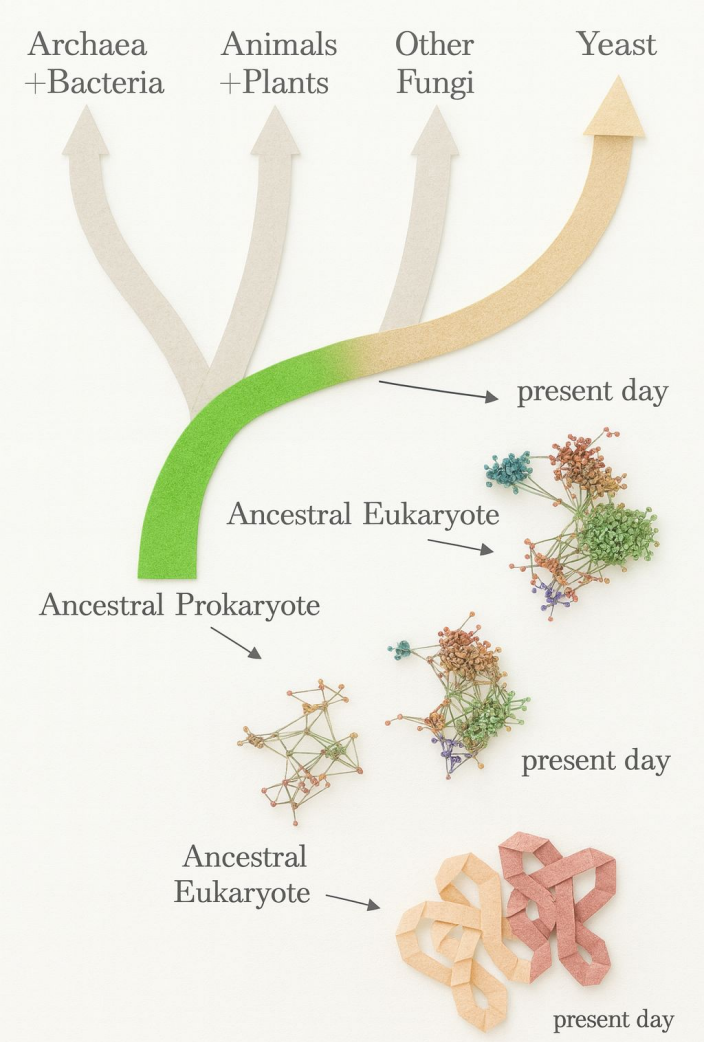


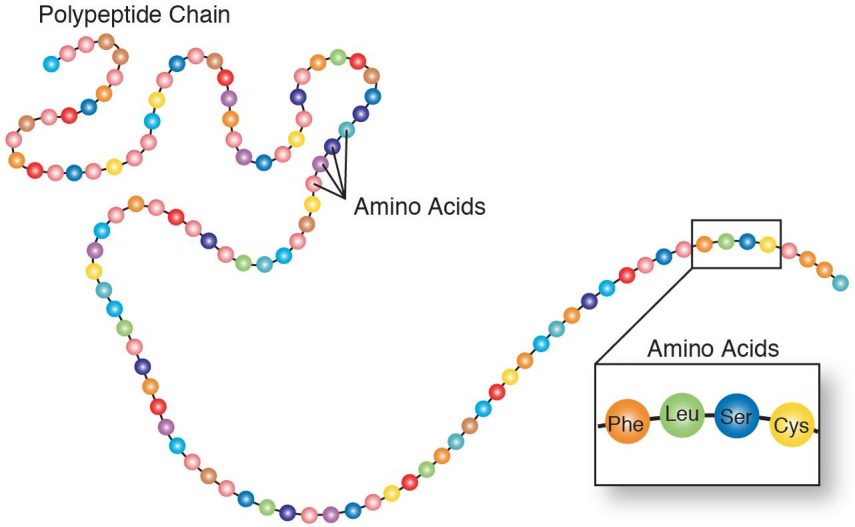
1. Proteins can fold in a astronomical number of different ways
2. Protein always finds a stable minimum of the folding landscape
3. Sequential process along a biased energy *funnel*

Levinthal's paradox

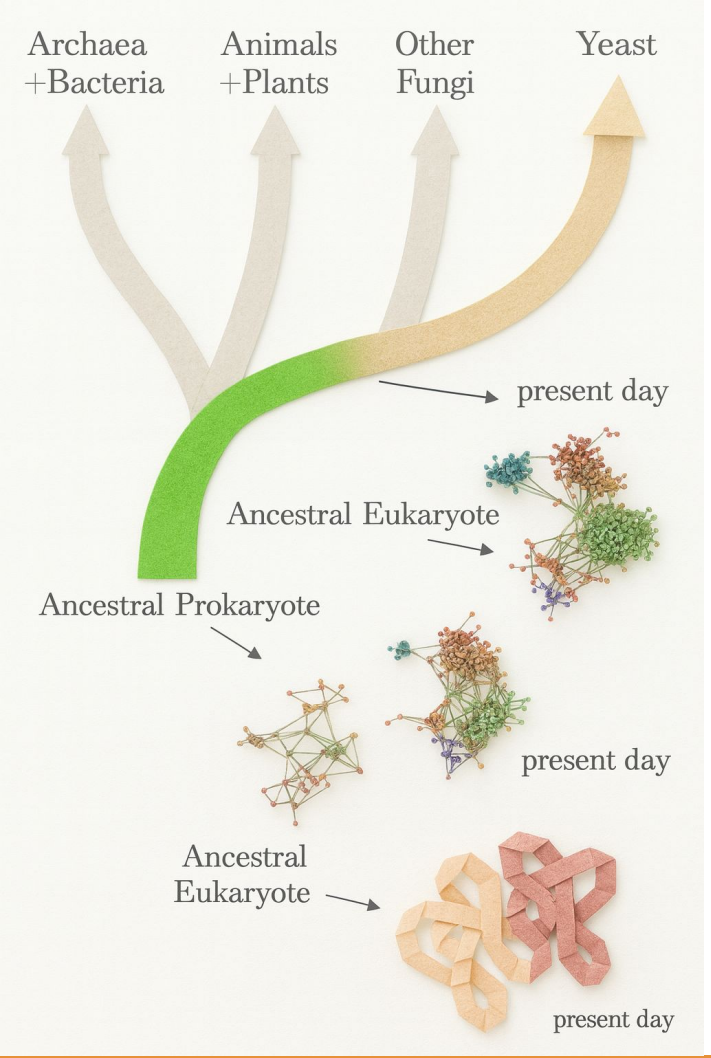


1. Proteins can fold in a astronomical number of different ways
2. Protein always finds a stable minimum of the folding landscape
3. Sequential process along a biased energy *funnel*


Archaea
+Bacteria
Animals
+Plants
Other
Fungi
Yeast
present day
Ancestral Eukaryote
Ancestral Prokaryote
present day
Ancestral
Eukaryote
present day


Polypeptide Chain
Amino Acids
Amino Acids
Phe
Leu
Ser
Cys

Multiple Sequence Alignment




```
                                                  *    .    :    .    .    *    : : :    .
Q5E940_BOVIN -----------MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMQQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE   76
  RLA0_HUMAN -----------MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMQQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE   76
  RLA0_MOUSE -----------MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMQQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE   76
    RLA0_RAT -----------MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMQQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE   76
  RLA0_CHICK -----------MPREDRATWKSNYFMKIIQLLDDYPKCFVVGADNVGSKQMQQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE   76
  RLA0_RANSY -----------MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMQQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--SALE   76
Q7ZUG3_BRARE -----------MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMQTIRLSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE   76
  RLA0_ICTPU -----------MPREDRATWKSNYFLKIIQLLNDYPKCFIVGADNVGSKQMQTIRLSLRGK-AIVLMGKNTMMRKAIRGHLENN--PALE   76
  RLA0_DROME -----------MVRENKAAWKAQYFIKVVELFDEFPKCFIVGADNVGSKQMQNIRTSIRGL-AVVLMGKNTMMRKAIRGHLENN--PQLE   76
  RLA0_DICDI -----------MSGAG-SKRKKLFIEKATKLFTTYDKMIVAEADFVGSSQLQKIRKSIRGI-GAVLMGKKTMIRKVIRDLADSK--PELD   75
Q54LP0_DICDI -----------MSGAG-SKRKNVFIEKATKLFTTYDKMIVAEADFVGSSQLQKIRKSIRGI-GAVLMGKKTMIRKVIRDLADSK--PELD   75
  RLA0_PLAF8 -----------MAKLSKQQKKQMYIEKLSSLIQQYSKILIVHVDNVGSNQMASVRKSLRGK-ATILMGKNTRIRTALKKNLQAV--PQIE   76
  RLA0_SULAC -----MIGLAVTTTKKIAKWKVDEVAELTEKLKTHKTIIIANIEGFPADKLHEIRKKLRGK-ADIKVTKNNLFNIALKNAG-----YDTK   79
  RLA0_SULTO ----MRIMAVITQERKIAKWKIEEVKELEQKLREYHTIIIANIEGFPADKLHDIRKKMRGM-AEIKVTKNTLFGIAAKNAG-----LDVS   80
  RLA0_SULSO ----MKRLALALKQRKVASWKLEEVKELTELIKNSNTILIGNLEGFPADKLHEIRKKLRGK-ATIKVTKNTLFKIAAKNAG-----IDIE   80
  RLA0_AERPE MSVVSLVGQMYKREKPIPEWKTLMLRELEELFSKHRVVLFADLTGTPTFVVQRVRKKLWKK-YPMMVAKKRILLRAMKAAGLE---LDDN   86
  RLA0_PYRAE -MMLAIGKRRYVRTRQYPARKVKIVSEATELLQKYPYVFLFDLHGLSNRILHEYRYRLRRY-GVIKIIKPTLFKIAFTKVYGG---IPAE   85
  RLA0_METAC ------MAEERHHTEHIPQWKKDEIENIKELIQSHKVFGMVGIEGILATKMQKIRRDLKDV-AVLKVSRNTLTERALNQLG-----ETIP   78
  RLA0_METMA ------MAEERHHTEHIPQWKKDEIENIKELIQSHKVFGMVRIEGILATKIQKIRRDLKDV-AVLKVSRNTLTERALNQLG-----ESIP   78
  RLA0_ARCFU ------MAAVRGS---PPEYKVRAVEEIKRMISSKPVVAIVSFRNVPAGQMQKIRREFRGK-AEIKVVKNTLLERALDALG-----GDYL   75
  RLA0_METKA MAVKAKGQPPSGYEPKVAEWKRREVKELKELMDEYENVGLVDLEGIPAPQLQEIRAKLRERDTIIRMSRNTLMRIALEEKLDER--PELE   88
  RLA0_METTH ------------MAHVAEWKKKEVQELHDLIKGYEVVGIANLADIPARQLQKMRQTLRDS-ALIRMSKKTLISLALEKAGREL--ENVD   74
  RLA0_METTL -------MITAESEHKIAPWKIEEVNKLKELLKNGQIVALVDMMEVPARQLQEIRDKIR-GTMTLKMSRNTLIERAIKEVAEETGNPEFA   82
  RLA0_METVA -------MIDAKSEHKIAPWKIEEVNALKELLKSANVIALIDMMEVPAVQLQEIRDKIR-DQMILKMSRNTLIKRAVEEVAEETGNPEFA   82
  RLA0_METJA --------METKVKAHVAPWKIEEVKTLKGLIKSKPVVAIVDMMDVPAPQLQEIRDKIR-DKVKLRMSRNTLIIRALKEAAEELNNPKLA   81
  RLA0_PYRAB ------------MAHVAEWKKKEVEELANLIKSYPVIALVDVSSMPAYPLSQMRRLIRENGGLLRVSRNTLIELAIKKAAQELGKPELE   77
  RLA0_PYRHO ------------MAHVAEWKKKEVEELAKLIKSYPVIALVDVSSMPAYPLSQMRRLIRENGGLLRVSRNTLIELAIKKAAKELGKPELE   77
  RLA0_PYRFU ------------MAHVAEWKKKEVEELANLIKSYPVVALVDVSSMPAYPLSQMRRLIRENNGLLRVSRNTLIELAIKKVAQELGKPELE   77
  RLA0_PYRKO ------------MAHVAEWKKKEVEELANIIKSYPVIALVDVAGVPAYPLSKMRDKLR-GKALLRVSRNTLIELAIKRAAQELGQPELE   76
  RLA0_HALMA -----MSAESERKTETIPEWKQEEVDAIVEMIESYESVGVVNIAGIPSRQLQDMRRDLHGT-AELRVSRNTLLERALDDVD-----DGLE   79
  RLA0_HALVO -----MSESEVRQTEVIPQWKREEVDELVDFIESYESVGVVGVAGIPSRQLQSMRRELHGS-AAVRMSRNTLVNRALDEVN-----DGFE   79
  RLA0_HALSA -----MSAEEQRTTEEVPEWKRQEVAELVDLIETYDSVGVVNVTGIPSKQLQDMRRGLHGQ-AALRMSRNTLLVRALEEAG-----DGLD   79
  RLA0_THEAC ------------MKEVSQQKKELVNEITQRIKASRSVAIVDFAGIRTRQIQDIRGKNRGK-INLKVIKKTLLFKALENLGD----EKLS   72
  RLA0_THEVO ------------MRKINPKKKEIVSELAQDITKSKAVAIVDIKGVRIRQMQDIKAKNRDK-VKIKVVKKTLLFKALDSIND----EKLT   72
  RLA0_PICTO ------------MTEPAQWKIDFVKNLENEINSRKVAAIVSIKGLRNNEFQKIRNSIRDK-ARIKVSRARLLRLAIENIGK----NNIV   72
       ruler 1.......10........20........30........40........50........60........70........80........90
```

SURF

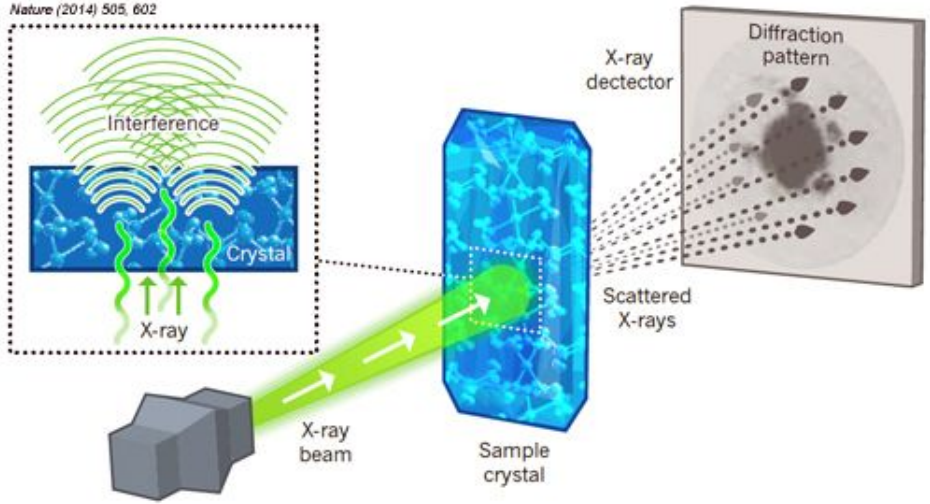
Guide to Obtain the Structure

1. X-Ray Crystallography

Gold Standard

Problems:

- crystallize the proteins
- Requires a Synchrotron
- Synchrotron usage: 150k
- Multi-year Project per protein
- Some proteins don’t crystallize

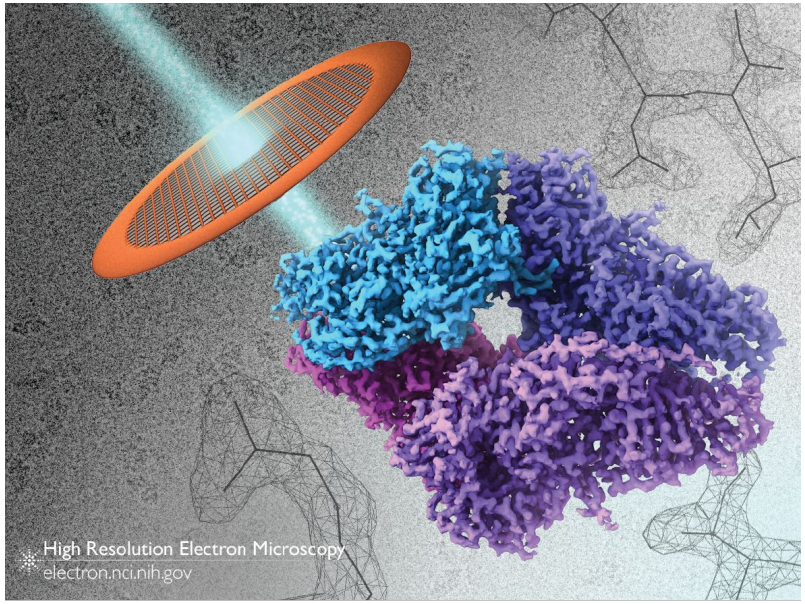


Guide to Obtain the Structure

2. Cryo-EM

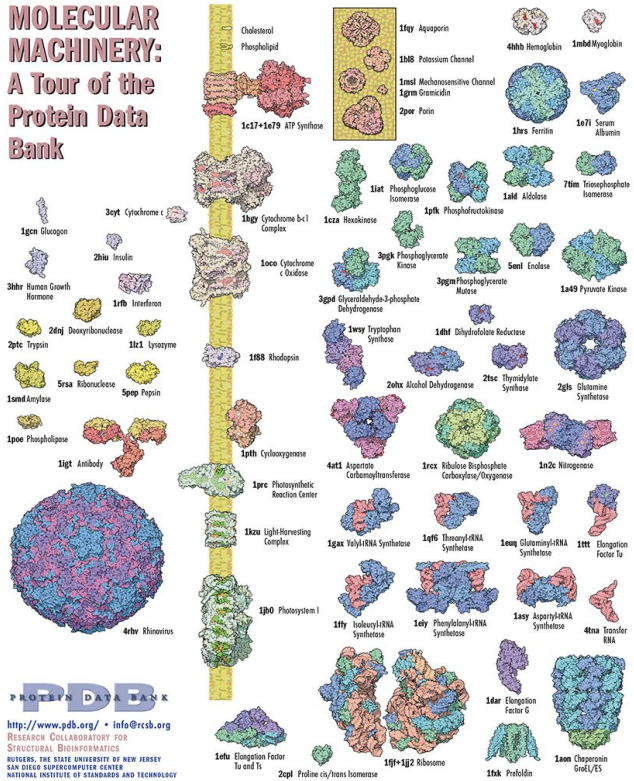
Problems:

- Expensive (5-10m)
- Maintenance
- 5-20k a day operation costs
- Not so good for small proteins





Protein Data Bank (PDB)



RCSB PDB Deposit Search Visualize Analyze Download Learn About Careers COVID-19 Help Contact us MyPDB

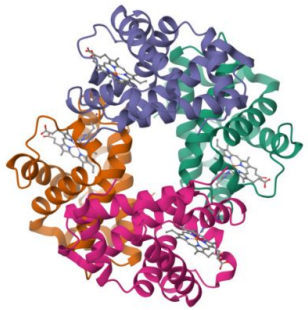
PDB PROTEIN DATA BANK — 241,055 Structures from the PDB archive — 1,068,577 Computed Structure Models (CSM)

Enter search term(s), Ligand ID or sequence — Include CSM — Advanced Search | Browse Annotations — Help

PDB-101 wwPDB EMDataResource NAKB wwPDB Foundation PDB-IHM

Structure Summary | Structure | Annotations | Experiment | Sequence | Genome | Ligands | Versions

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (HEM)

Global Symmetry: Cyclic - C2 (Explore in 3D)
Global Stoichiometry: Hetero 4-mer - A2B2

Pseudo Symmetry: Dihedral - D2 (Explore in 3D)
Pseudo Stoichiometry: Homo 4-mer - A4

Find Similar Assemblies

Biological assembly 1 assigned by authors and generated by PISA (software)

Macromolecule Content

- Total Structure Weight: 64.55 kDa
- Atom Count: 4,997
- Modeled Residue Count: 572
- Deposited Residue Count: 574
- Unique protein chains: 2

Display Files | Download Files | Data API

1A3N | pdb_00001a3n

DEOXY HUMAN HEMOGLOBIN

PDB DOI: https://doi.org/10.2210/pdb1A3N/pdb

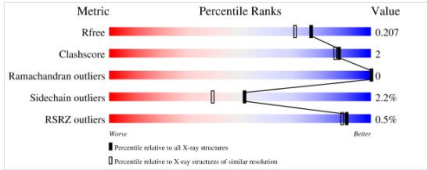
Classification: OXYGEN TRANSPORT
Organism(s): Homo sapiens
Mutation(s): No

Deposited: 1998-01-22 **Released:** 1998-04-29
Deposition Author(s): Tame, J., Vallone, B.

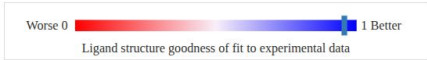
Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 1.80 Å
R-Value Free:
0.220 (Depositor), 0.210 (DCC)
R-Value Work:
0.171 (Depositor), 0.160 (DCC)

wwPDB Validation — 3D Report | Full Report



Ligand Structure Quality Assessment



This is version 1.3 of the entry. See complete history.

Literature — Download Primary Citation

The structures of deoxy human haemoglobin and the mutant Hb Tyralpha42His at 120 K.

Tame, J.R., Vallone, B.
(2000) Acta Crystallogr D Biol Crystallogr **56**: 805-811

PubMed: 10930827 Search on PubMed
DOI: https://doi.org/10.1107/s0907444900006387
Primary Citation of Related Structures:
1A3N, 1A3O

Open

rcsb_pdb_1A3N.fasta
~/Documents/projects

```
>1A3N_1|Chains A, C|HEMOGLOBIN (ALPHA CHAIN)|Homo sapiens (9606)
VLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR
>1A3N_2|Chains B, D|HEMOGLOBIN (BETA CHAIN)|Homo sapiens (9606)
VHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH
```

Open

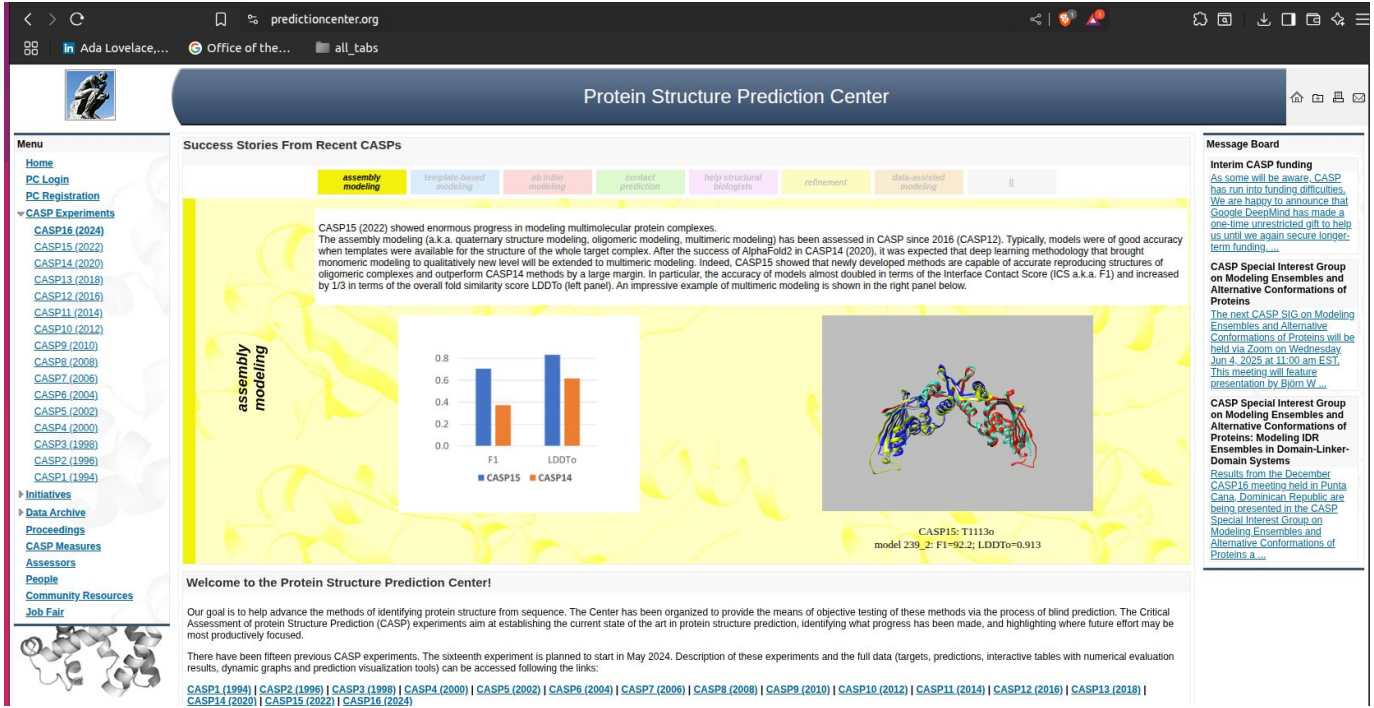
1a3n.cif
~/Documents/projects

```
_atom_site.type_symbol
_atom_site.label_atom_id
_atom_site.label_alt_id
_atom_site.label_comp_id
_atom_site.label_asym_id
_atom_site.label_entity_id
_atom_site.label_seq_id
_atom_site.pdbx_PDB_ins_code
_atom_site.Cartn_x
_atom_site.Cartn_y
_atom_site.Cartn_z
_atom_site.occupancy
_atom_site.B_iso_or_equiv
_atom_site.pdbx_formal_charge
_atom_site.auth_seq_id
_atom_site.auth_comp_id
_atom_site.auth_asym_id
_atom_site.auth_atom_id
_atom_site.pdbx_PDB_model_num
ATOM   1    N  N   . VAL A 1 1   ? 10.720  19.523  6.163   1.00 21.36 ? 1   VAL A N   1
ATOM   2    C  CA  . VAL A 1 1   ? 10.228  20.761  6.807   1.00 24.26 ? 1   VAL A CA  1
ATOM   3    C  C   . VAL A 1 1   ? 8.705   20.714  6.878   1.00 18.62 ? 1   VAL A C   1
ATOM   4    O  O   . VAL A 1 1   ? 8.164   20.005  6.015   1.00 19.87 ? 1   VAL A O   1
ATOM   5    C  CB  . VAL A 1 1   ? 10.602  22.000  5.966   1.00 27.19 ? 1   VAL A CB  1
ATOM   6    C  CG1 . VAL A 1 1   ? 10.307  23.296  6.700   1.00 31.86 ? 1   VAL A CG1 1
ATOM   7    C  CG2 . VAL A 1 1   ? 12.065  21.951  5.544   1.00 31.74 ? 1   VAL A CG2 1
ATOM   8    N  N   . LEU A 1 2   ? 8.091   21.453  7.775   1.00 16.19 ? 2   LEU A N   1
ATOM   9    C  CA  . LEU A 1 2   ? 6.624   21.451  7.763   1.00 17.31 ? 2   LEU A CA  1
ATOM   10   C  C   . LEU A 1 2   ? 6.176   22.578  6.821   1.00 18.55 ? 2   LEU A C   1
ATOM   11   O  O   . LEU A 1 2   ? 6.567   23.730  7.022   1.00 18.72 ? 2   LEU A O   1
ATOM   12   C  CB  . LEU A 1 2   ? 6.020   21.707  9.129   1.00 18.34 ? 2   LEU A CB  1
ATOM   13   C  CG  . LEU A 1 2   ? 6.386   20.649  10.198  1.00 17.39 ? 2   LEU A CG  1
ATOM   14   C  CD1 . LEU A 1 2   ? 5.998   21.119  11.577  1.00 17.99 ? 2   LEU A CD1 1
ATOM   15   C  CD2 . LEU A 1 2   ? 5.730   19.337  9.795   1.00 16.96 ? 2   LEU A CD2 1
ATOM   16   N  N   . SER A 1 3   ? 5.380   22.237  5.852   1.00 15.02 ? 3   SER A N   1
```

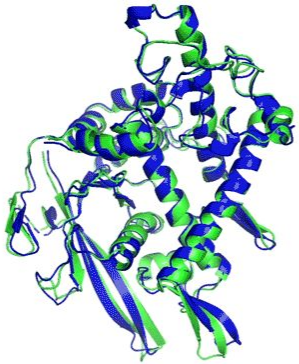
SURF

CASP Competition

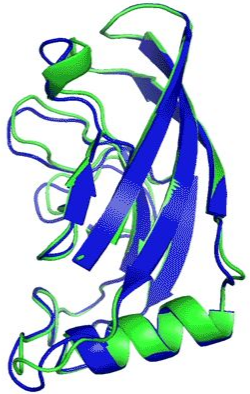
Critical Assessment of Structure Prediction



Local Distance Difference Test (LDDT)
Global Distance Test (GDT)



T1037 / 6vr4
90.7 GDT
(RNA polymerase domain)



T1049 / 6y4f
93.3 GDT
(adhesin tip)

Experimental result
Computational prediction

CASP data is double-blind, unpublished and X-ray Crystallography obtained

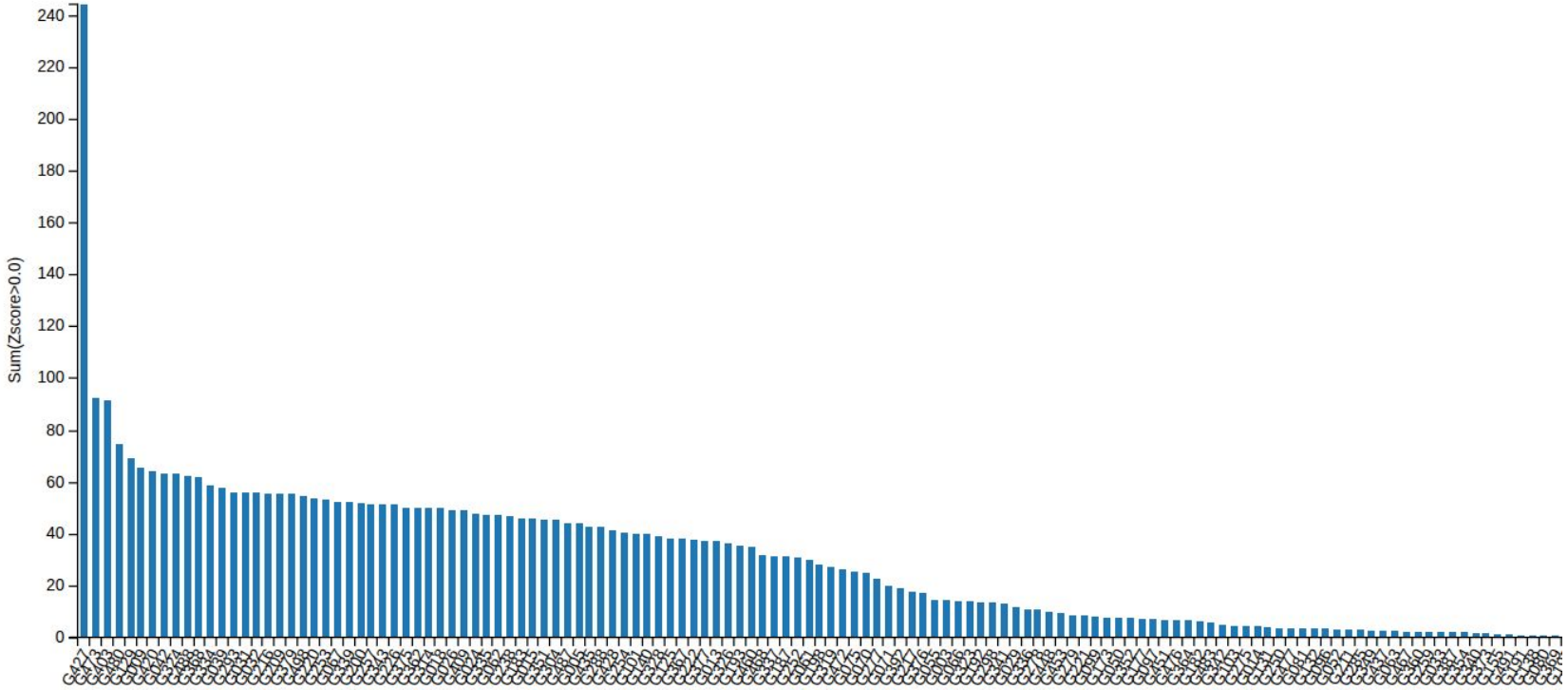
Hard and **expensive** to replicate

Solving Folding

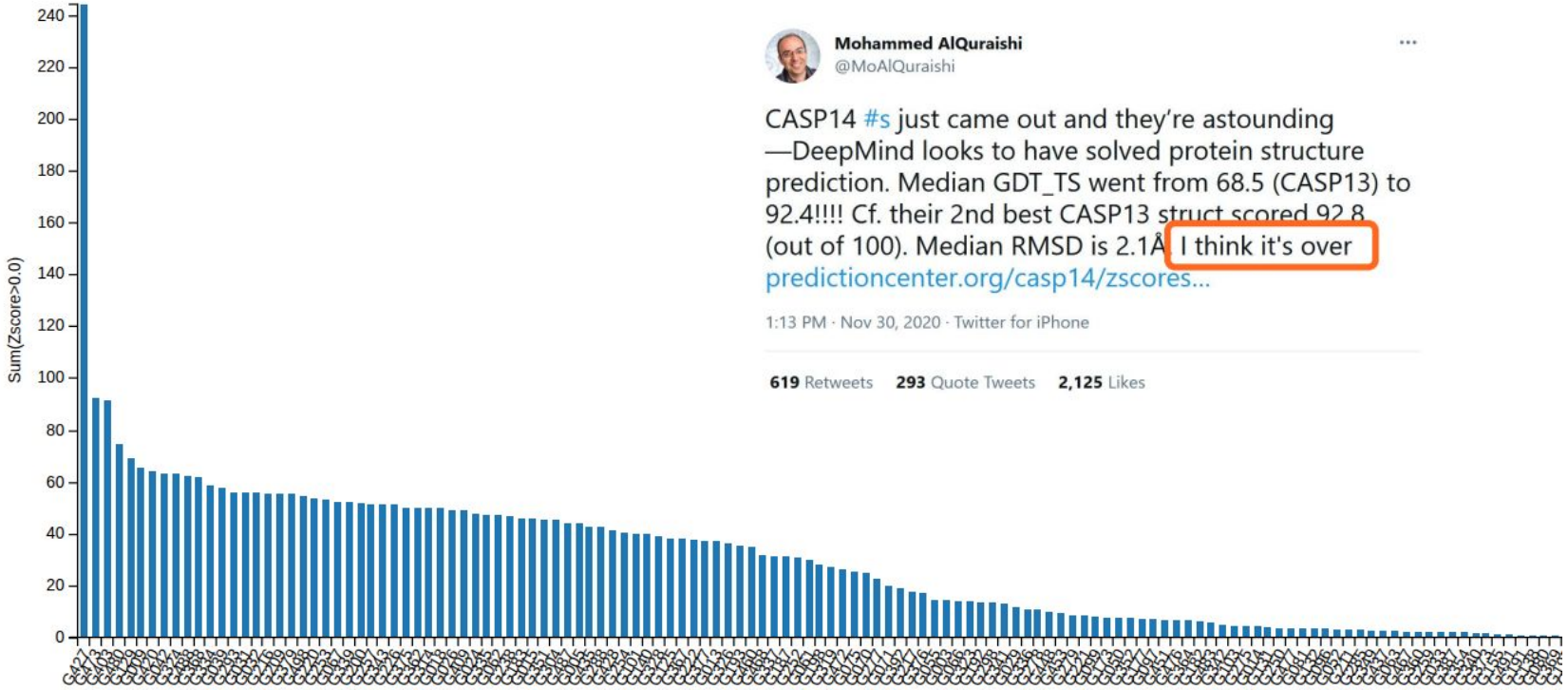


High Performance Machine Learning Group

Competition Results



Competition Results



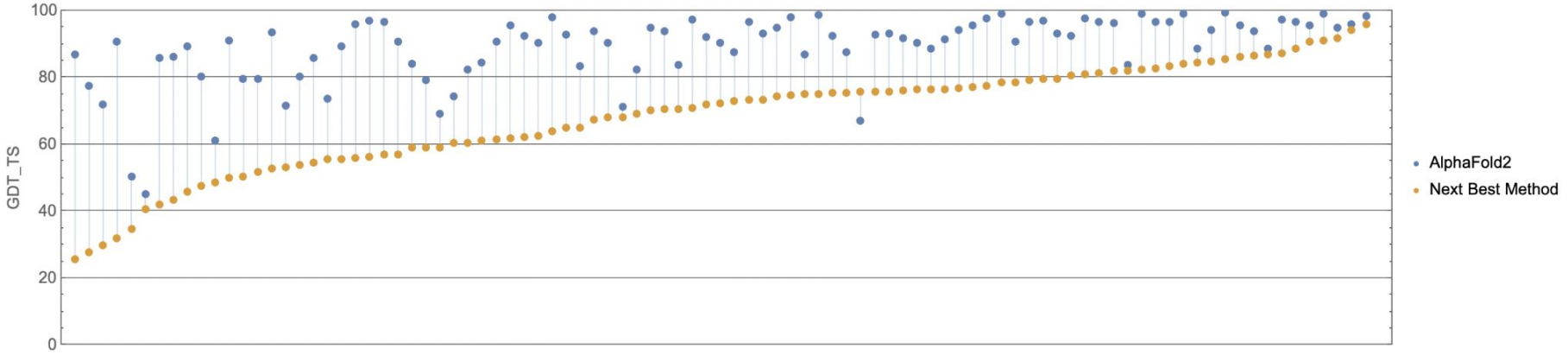
Mohammed AlQuraishi
@MoAlQuraishi

CASP14 #s just came out and they're astounding —DeepMind looks to have solved protein structure prediction. Median GDT_TS went from 68.5 (CASP13) to 92.4!!!! Cf. their 2nd best CASP13 struct scored 92.8 (out of 100). Median RMSD is 2.1Å. I think it's over predictioncenter.org/casp14/zscores...

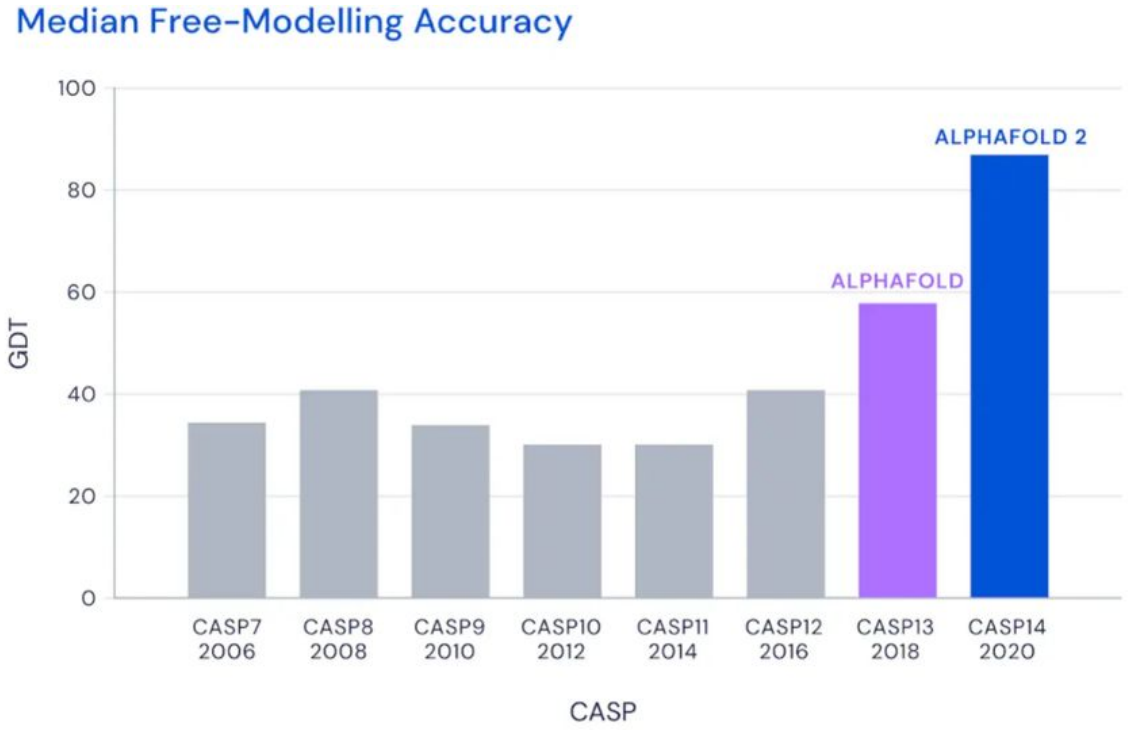
1:13 PM · Nov 30, 2020 · Twitter for iPhone

619 Retweets **293** Quote Tweets **2,125** Likes

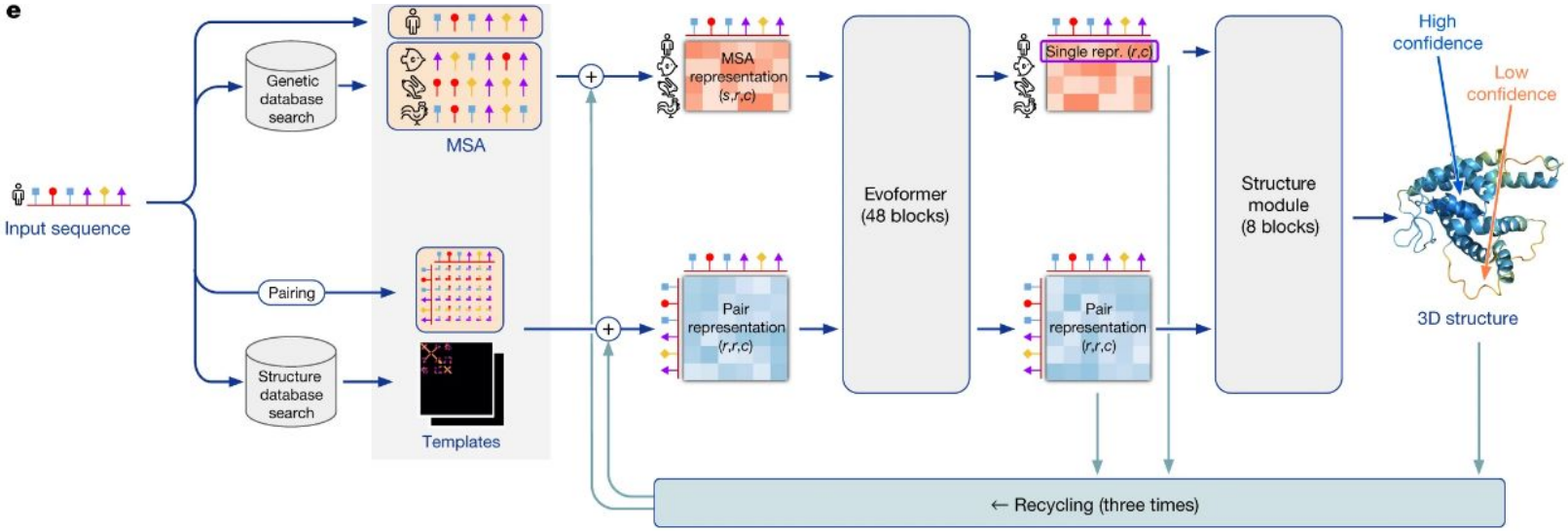
Competition Results



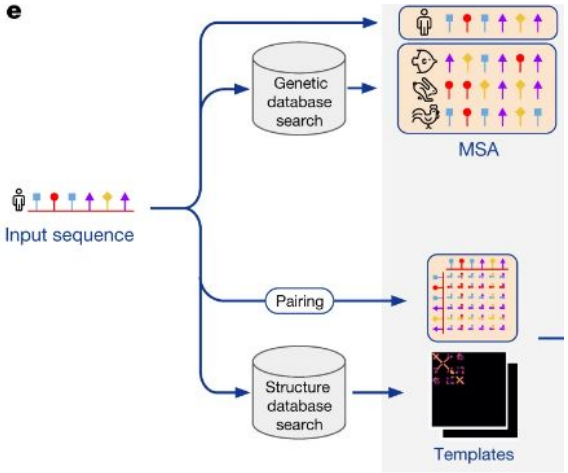
Competition Results



AlphaFold-2


e
Input sequence
Genetic database search
MSA
Pairing
Structure database search
Templates
MSA representation (s,r,c)
Pair representation (r,r,c)
Evoformer (48 blocks)
Single repr. (r,c)
Pair representation (r,r,c)
Structure module (8 blocks)
High confidence
Low confidence
3D structure
← Recycling (three times)

AlphaFold-2

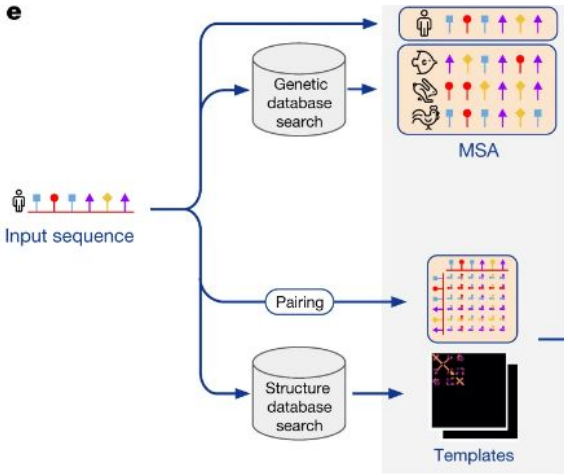


Multiple Sequence Alignment

Aligned Residues

Structural Templates for Sequence

AlphaFold-2



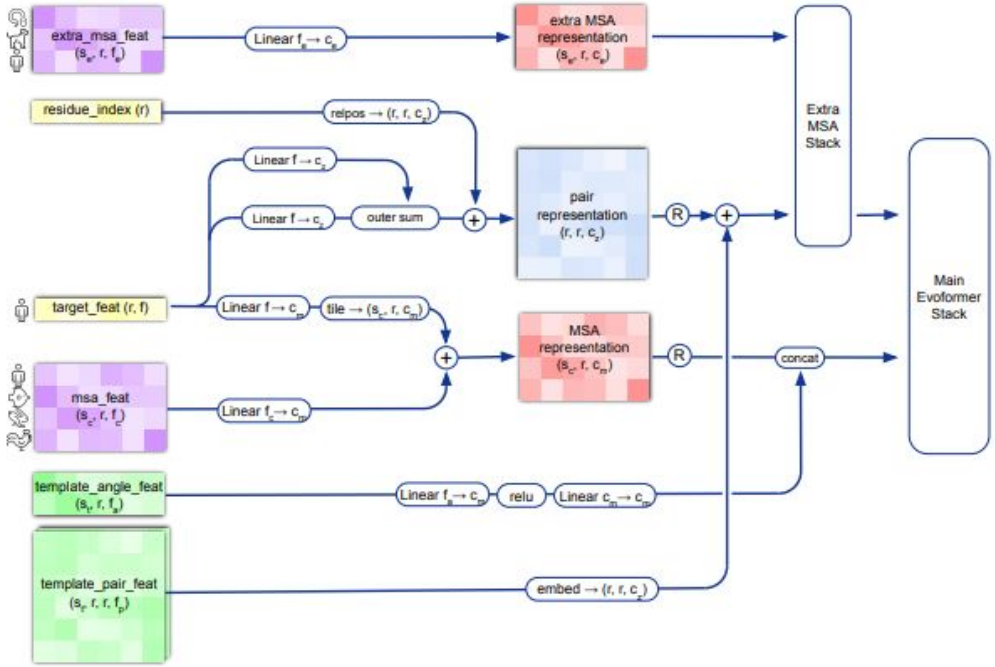
Search similar sequences and templates

1. Hhblits
2. JackHmmer

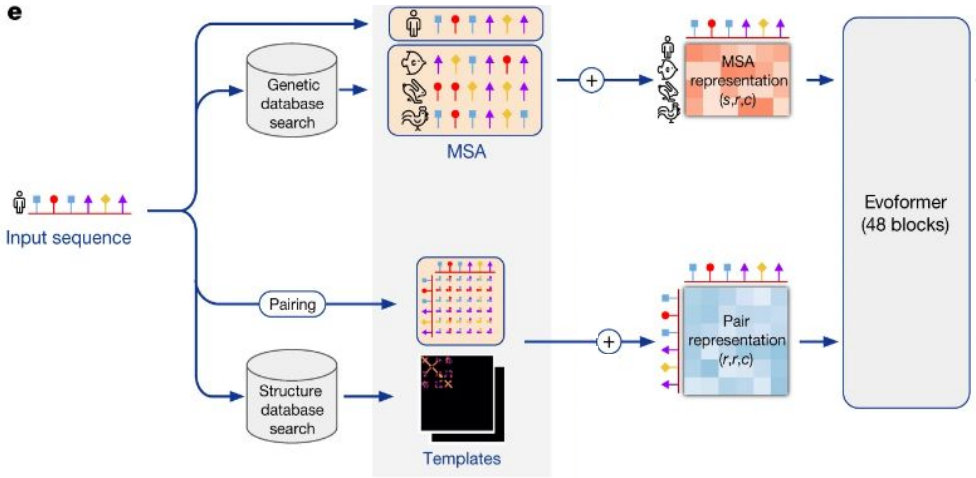
Search over **sequence** databases:

1. UniPro
2. MGnify
3. BFD

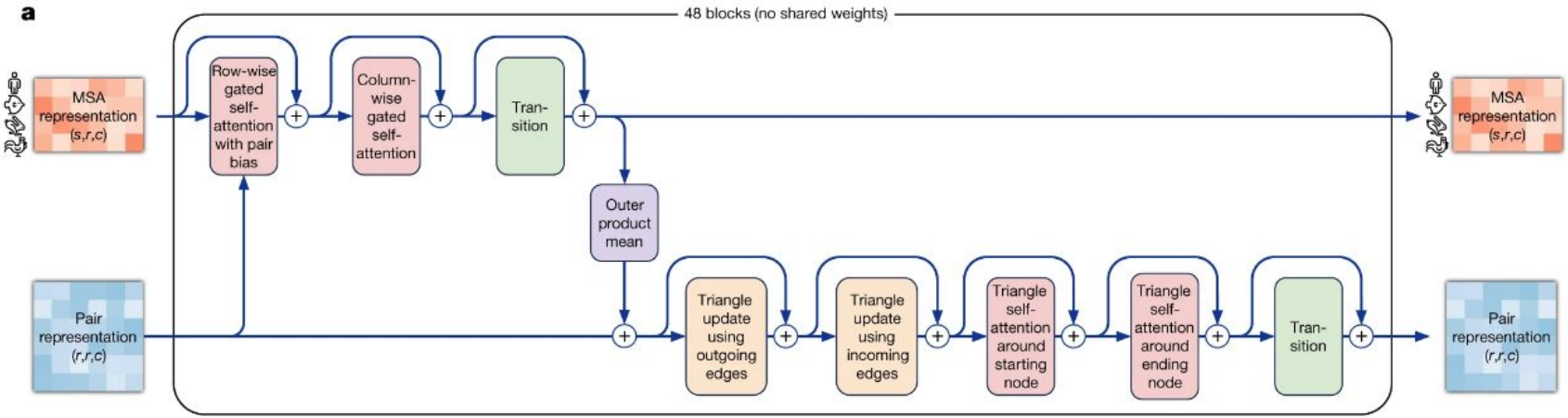
AlphaFold-2


extra_msa_feat
Linear
extra MSA representation
residue_index (r)
relpos
Linear
Linear
outer sum
pair representation
Extra MSA Stack
target_feat (r, f)
Linear
tile
MSA representation
R
concat
Main Evoformer Stack
msa_feat
Linear
template_angle_feat
Linear
relu
Linear
template_pair_feat
embed

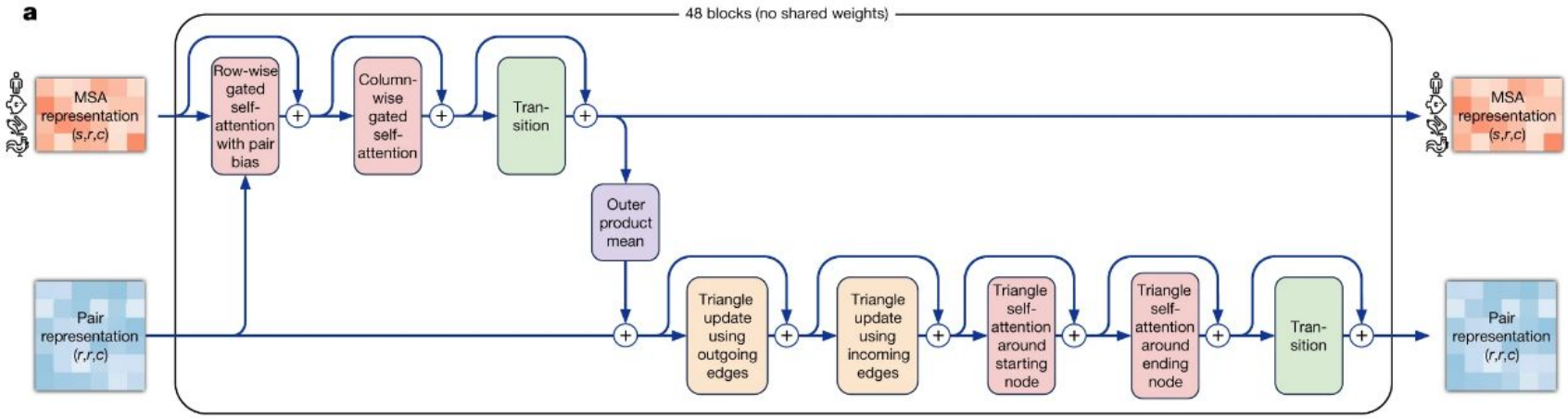
AlphaFold-2


e
Genetic database search
MSA
MSA representation (s,r,c)
Input sequence
Evoformer (48 blocks)
Pairing
Pair representation (r,r,c)
Structure database search
Templates

AlphaFold-2: EvoFormer



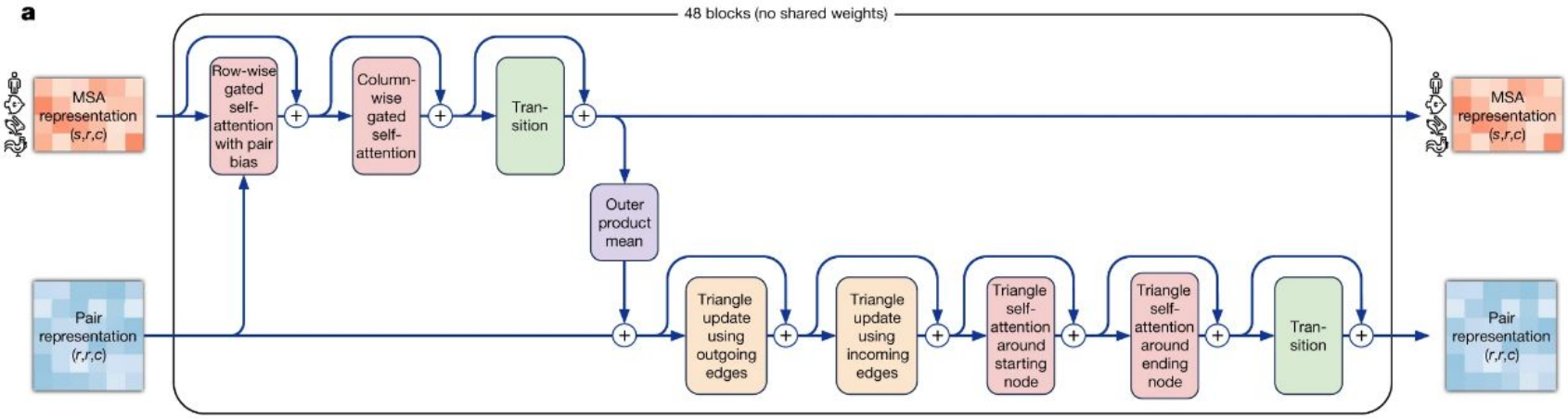
AlphaFold-2: EvoFormer

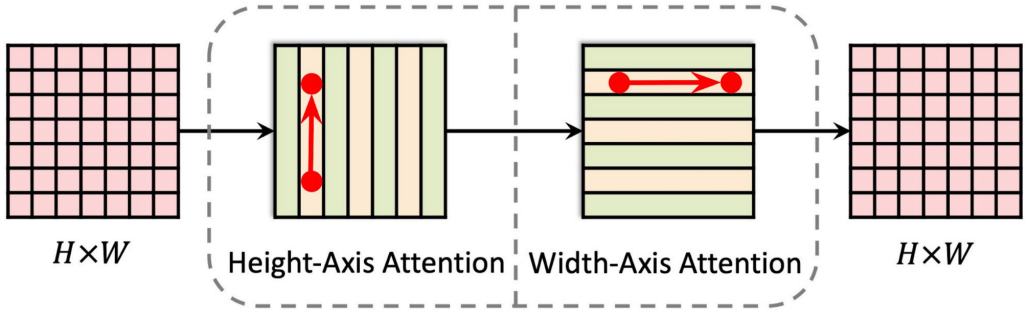


1. Extract information from similar sequences

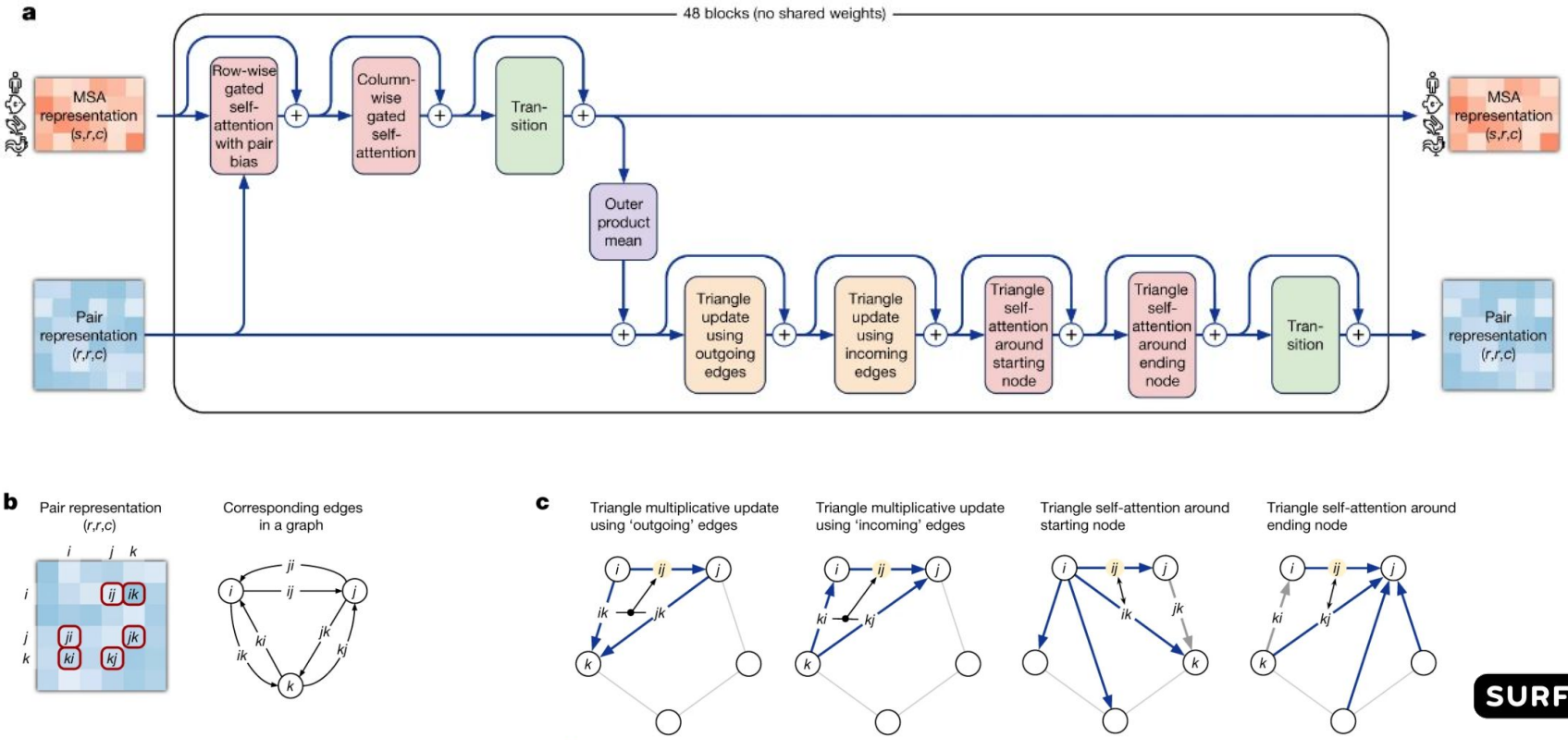
2. Learn spatial relations between amino acids

AlphaFold-2: EvoFormer

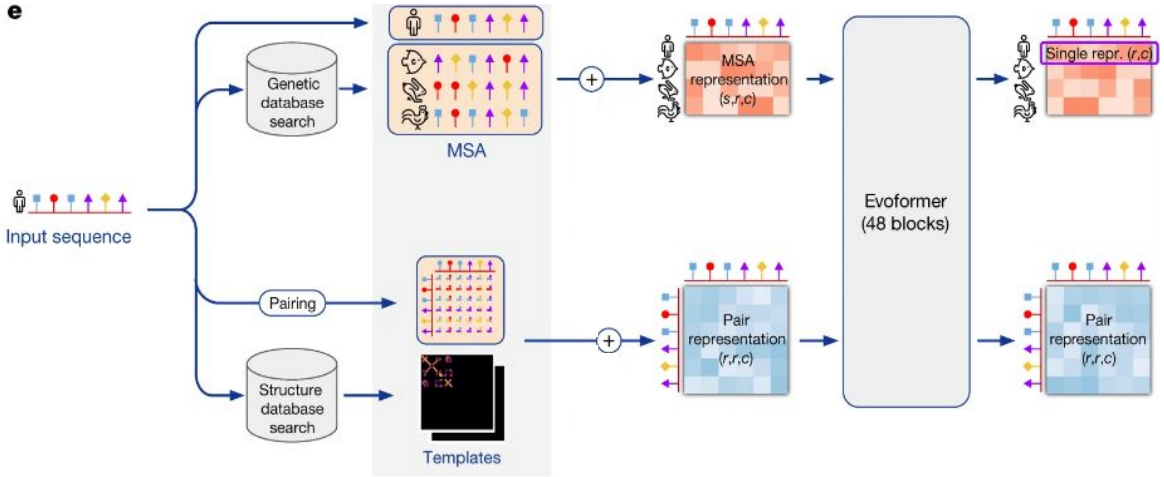




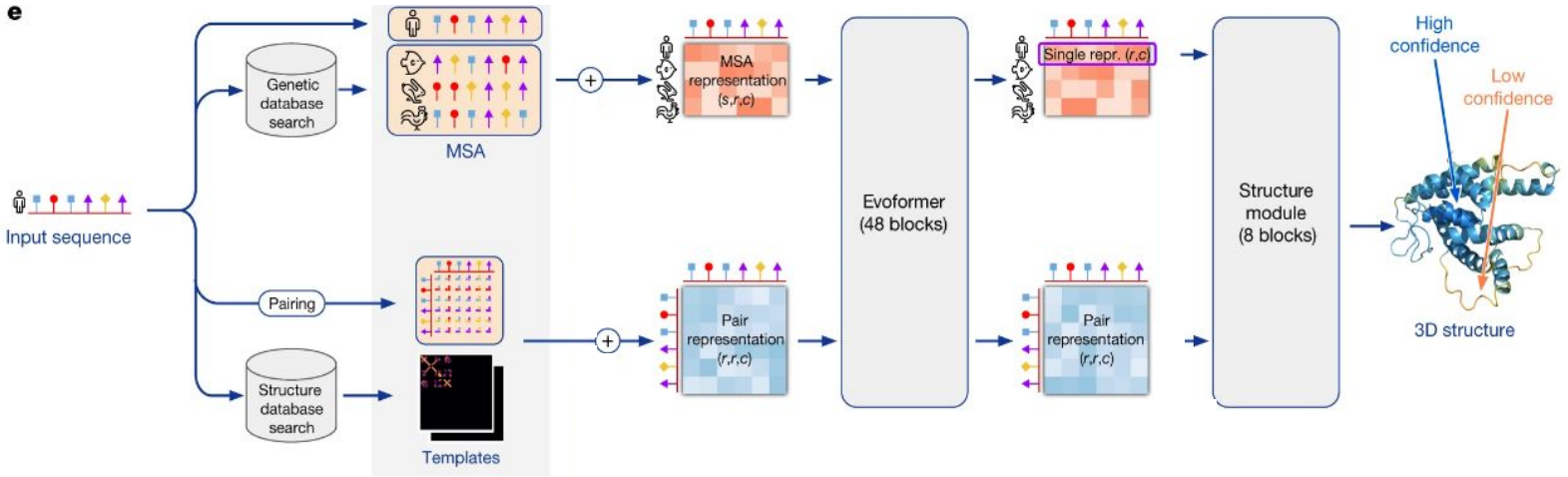
AlphaFold-2: Triangle Inequality



AlphaFold-2


e
Input sequence
Genetic database search
MSA
Pairing
Structure database search
Templates
MSA representation (s,r,c)
Pair representation (r,r,c)
Evoformer (48 blocks)
Single repr. (r,c)
Pair representation (r,r,c)

AlphaFold-2


e
Input sequence
Genetic database search
MSA
Pairing
Structure database search
Templates
MSA representation (s,r,c)
Pair representation (r,r,c)
Evoformer (48 blocks)
Single repr. (r,c)
Pair representation (r,r,c)
Structure module (8 blocks)
High confidence
Low confidence
3D structure

AlphaFold-2: Invariant Point Attention

From **representations** to concrete **3D coordinates**

Parameterize torsion angles for each amino-acid

Don’t learn rotations or translations in space

AlphaFold-2: Invariant Point Attention

From **representations** to concrete **3D coordinates**

Parameterize torsion angles for each amino-acid

Don’t learn rotations or translations in space

Structure module

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- **End-to-end folding** instead of gradient descent
- Protein backbone = gas of 3-D rigid bodies (chain is learned!)

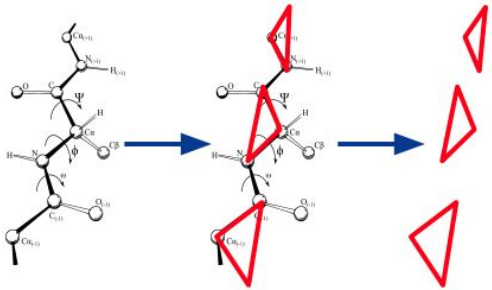
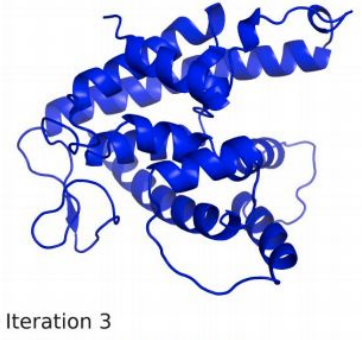


Image: Dcrjsr, vectorised Adam Rędzikowski (CC BY 3.0, Wikipedia)

- **3-D equivariant transformer architecture** updates the rigid bodies / backbone
  - Also builds the side chains

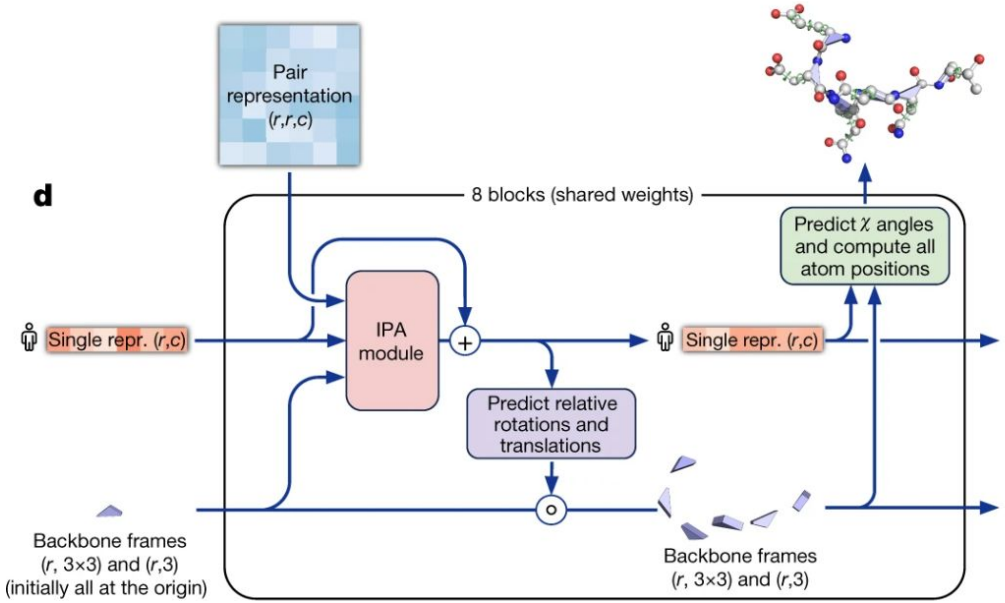


AlphaFold-2: Invariant Point Attention

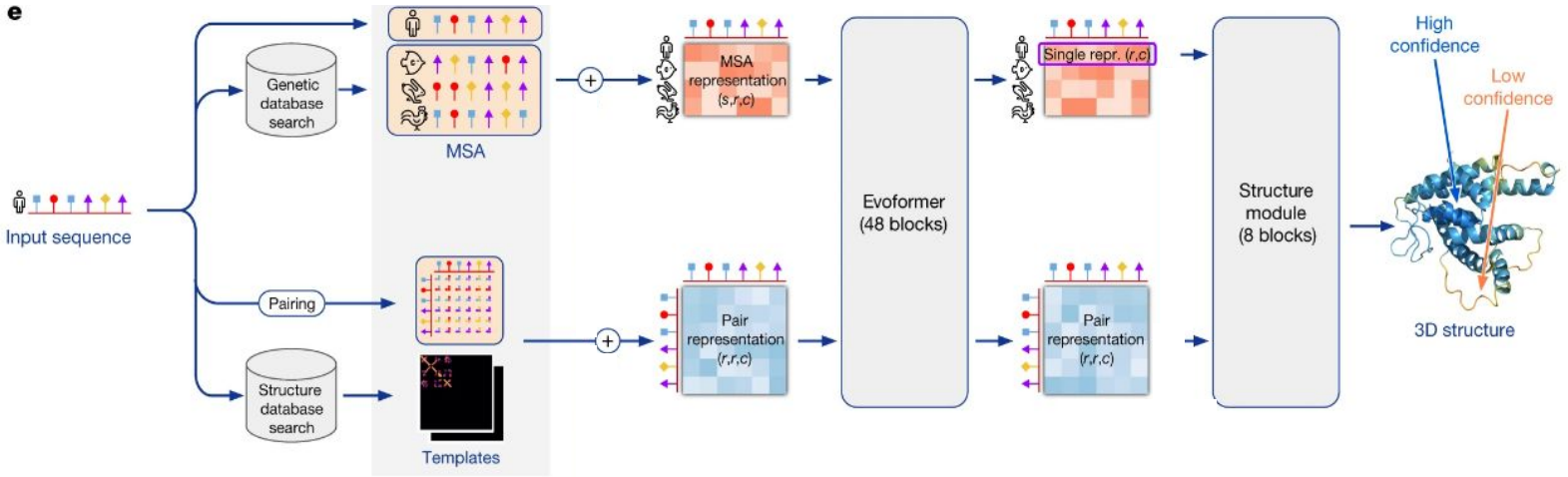
From **representations** to concrete **3D coordinates**

Parameterize torsion angles for each amino-acid

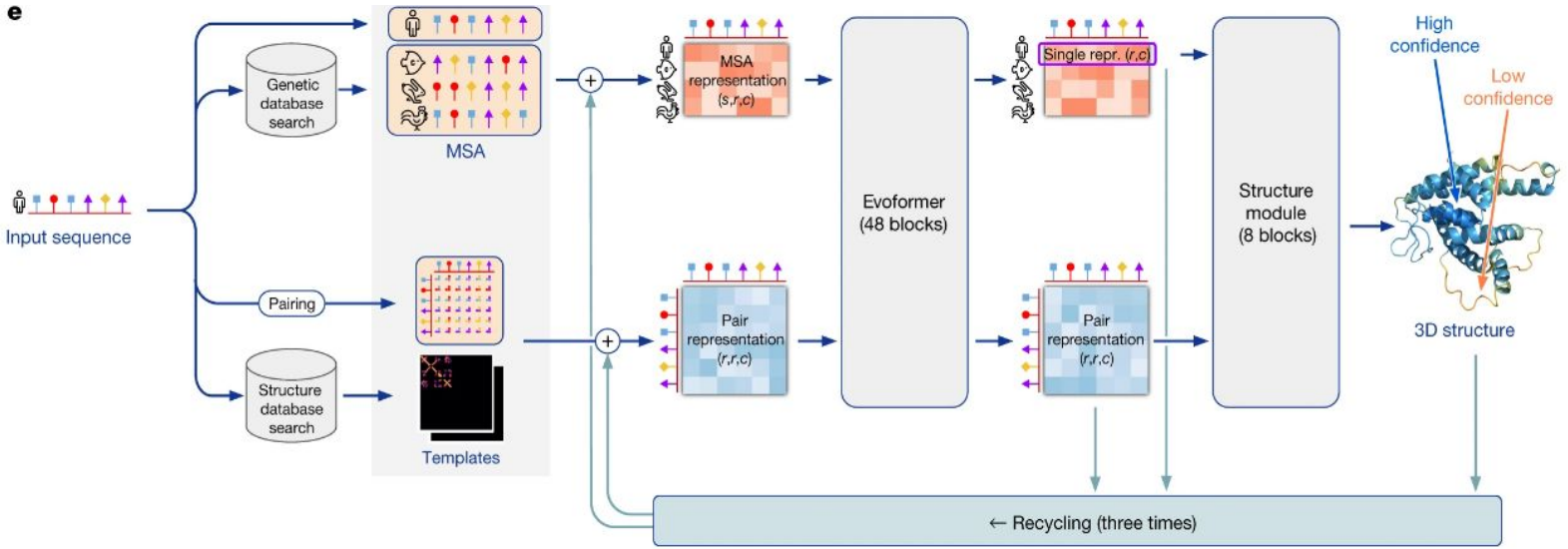
Don’t learn rotations or translations in space



AlphaFold-2


e
Input sequence
Genetic database search
MSA
Pairing
Structure database search
Templates
MSA representation (s,r,c)
Pair representation (r,r,c)
Evoformer (48 blocks)
Single repr. (r,c)
Pair representation (r,r,c)
Structure module (8 blocks)
High confidence
Low confidence
3D structure

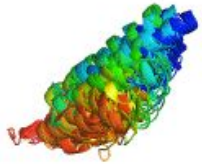
AlphaFold-2


e
Input sequence
Genetic database search
MSA
Pairing
Structure database search
Templates
MSA representation (s,r,c)
Pair representation (r,r,c)
Evoformer (48 blocks)
Single repr. (r,c)
Pair representation (r,r,c)
Structure module (8 blocks)
High confidence
Low confidence
3D structure
← Recycling (three times)

AlphaFold-2

List of *tricks* to make AF2 work

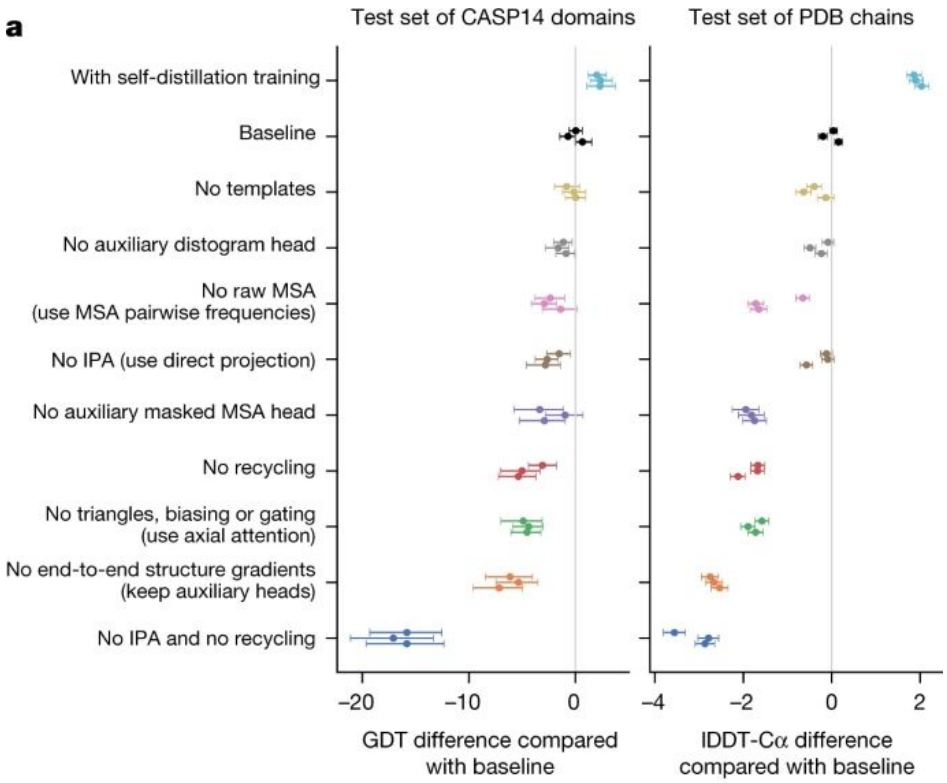
1. Filtering of Data
2. self-distillation
3. Ensembling multiple models
4. Recurrence
5. Auxiliary losses (MLP heads) for regularization
6. AMBER Relaxation
7. Masked language modeling training
8. Model confidence predictions (plddt and ptm)
9. Gradient Checkpointing



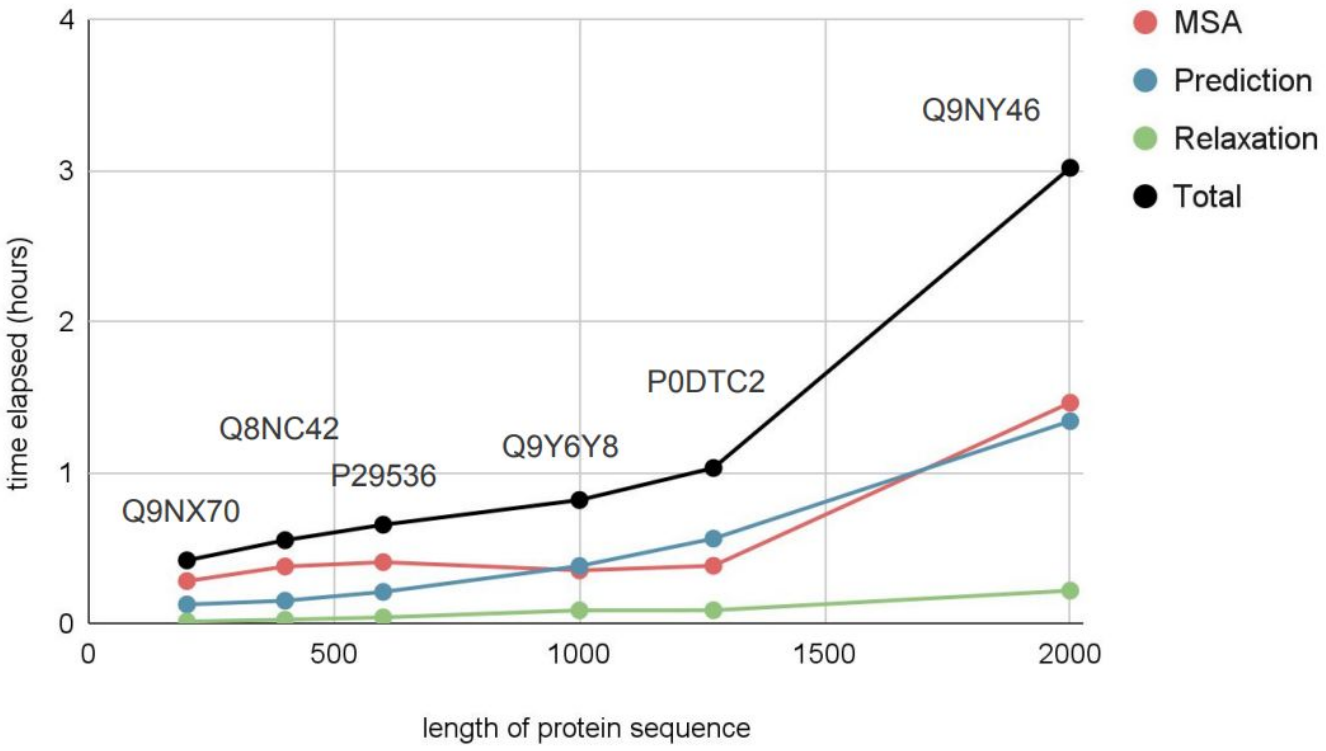
1

$$\mathcal{L} = \begin{cases} 0.5\mathcal{L}_{\text{FAPE}} + 0.5\mathcal{L}_{\text{aux}} + 0.3\mathcal{L}_{\text{dist}} + 2.0\mathcal{L}_{\text{msa}} + 0.01\mathcal{L}_{\text{conf}} & \text{training} \\ 0.5\mathcal{L}_{\text{FAPE}} + 0.5\mathcal{L}_{\text{aux}} + 0.3\mathcal{L}_{\text{dist}} + 2.0\mathcal{L}_{\text{msa}} + 0.01\mathcal{L}_{\text{conf}} + 0.01\mathcal{L}_{\text{exp resolved}} + 1.0\mathcal{L}_{\text{viol}} & \text{fine-tuning} \end{cases},$$

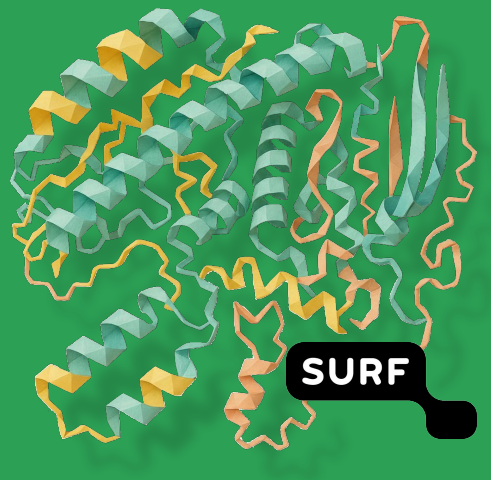
AlphaFold-2: prediction time (A100)



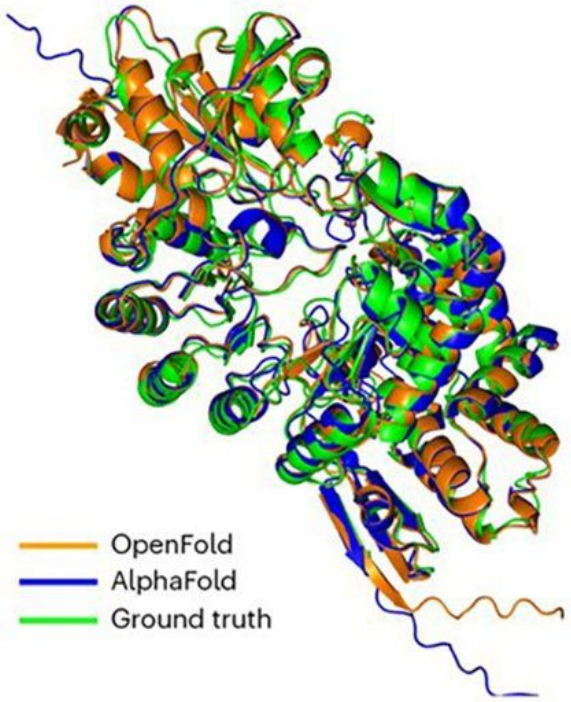
AlphaFold-2: prediction time (A100)



Suite of Models

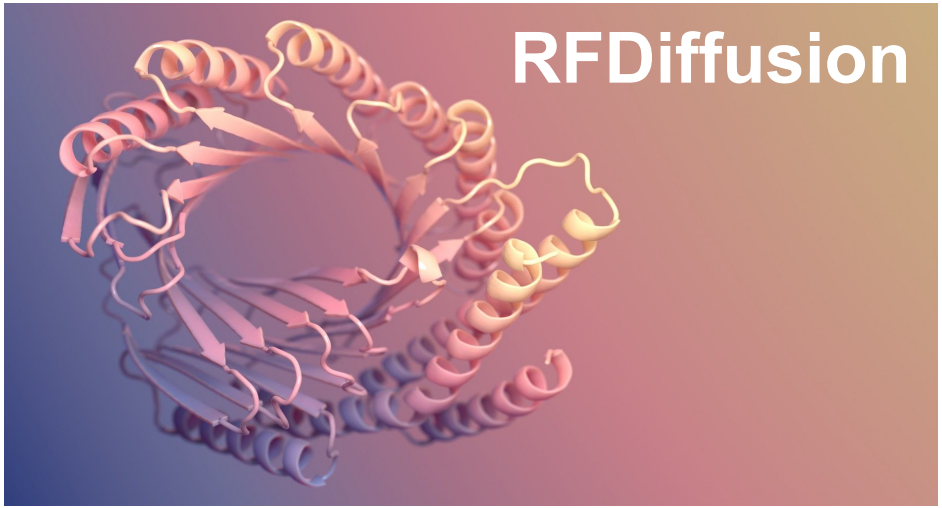


OpenFold

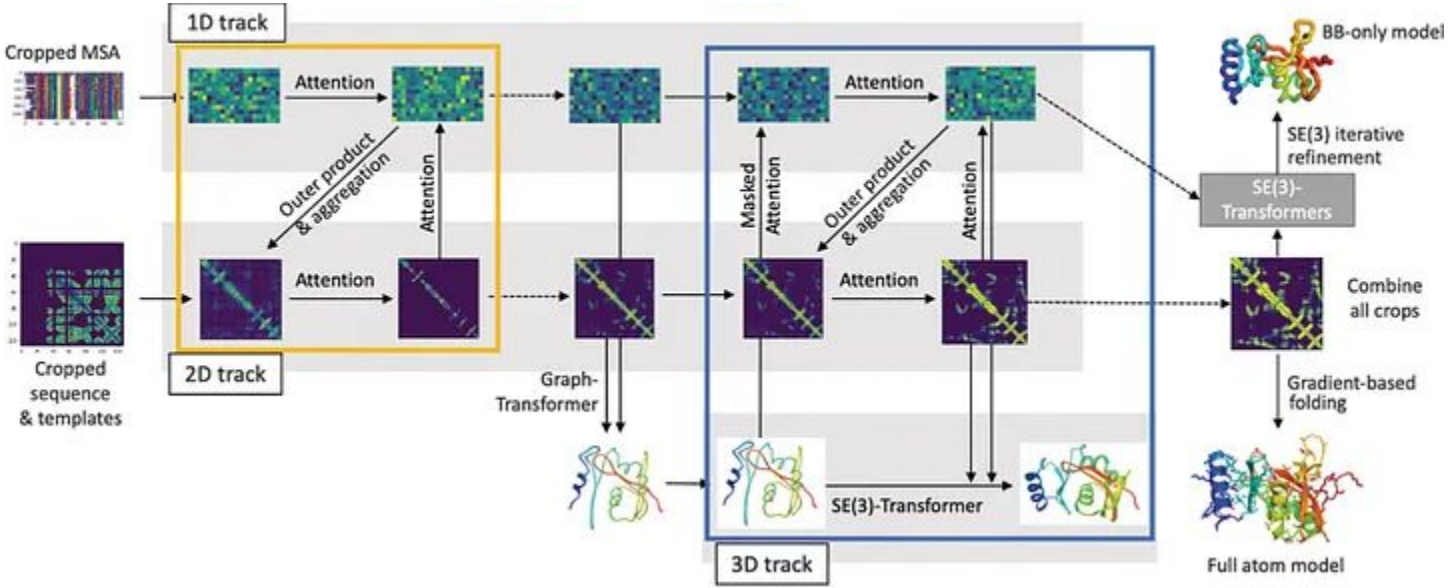


RosettaFold

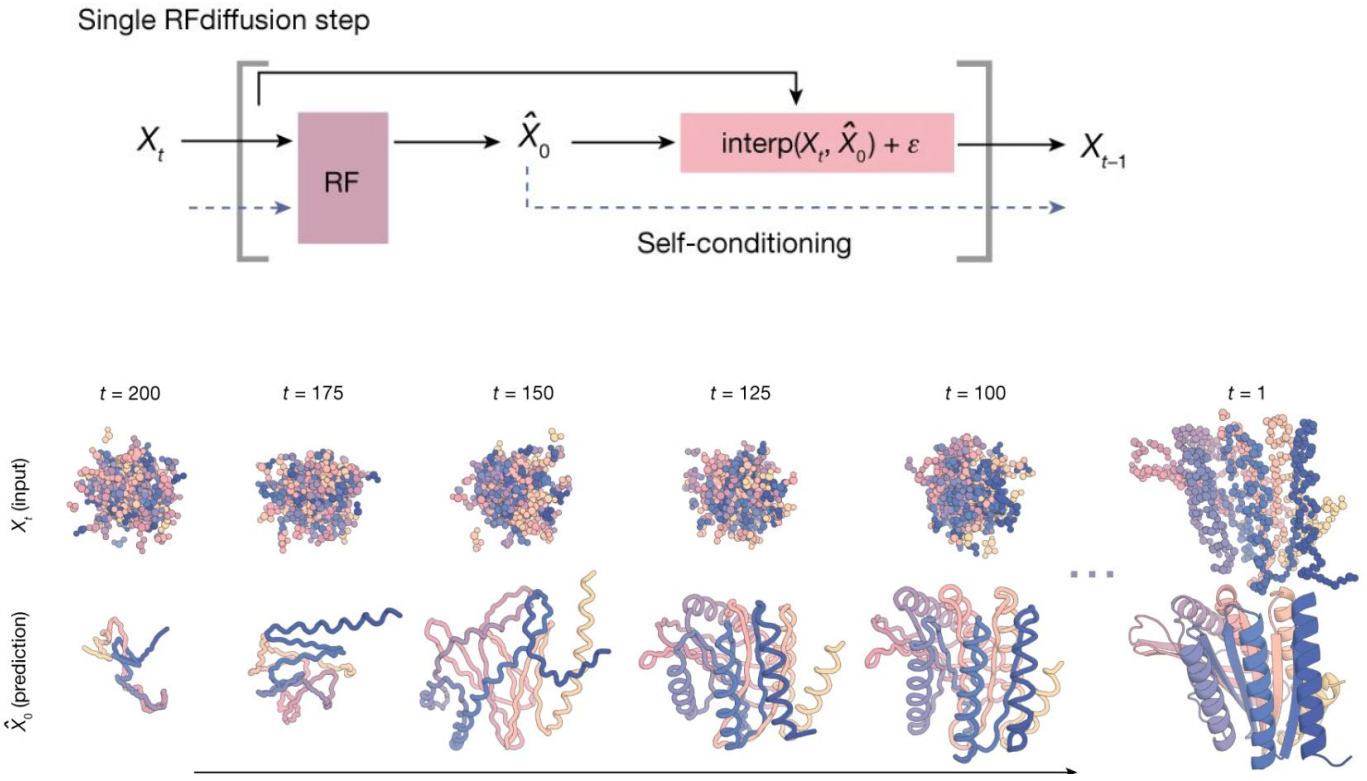




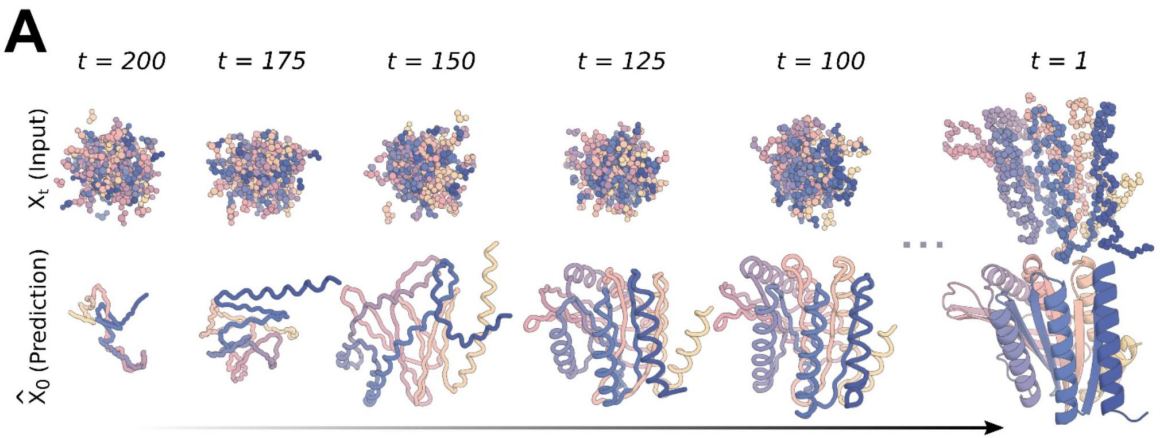
RosettaFold

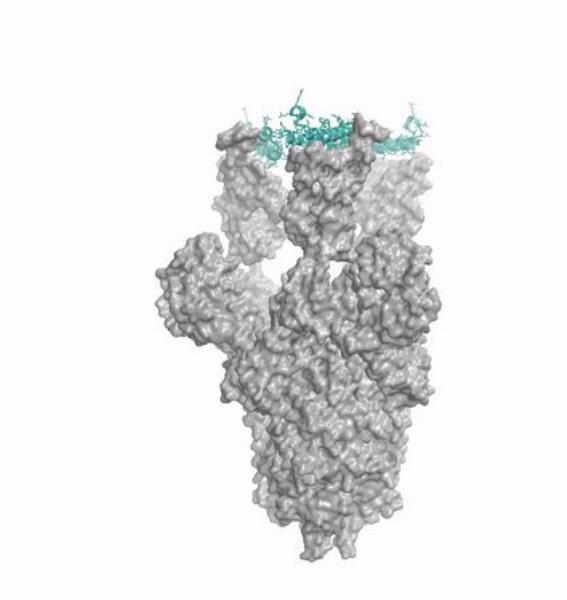


RosettaFold Diffusion



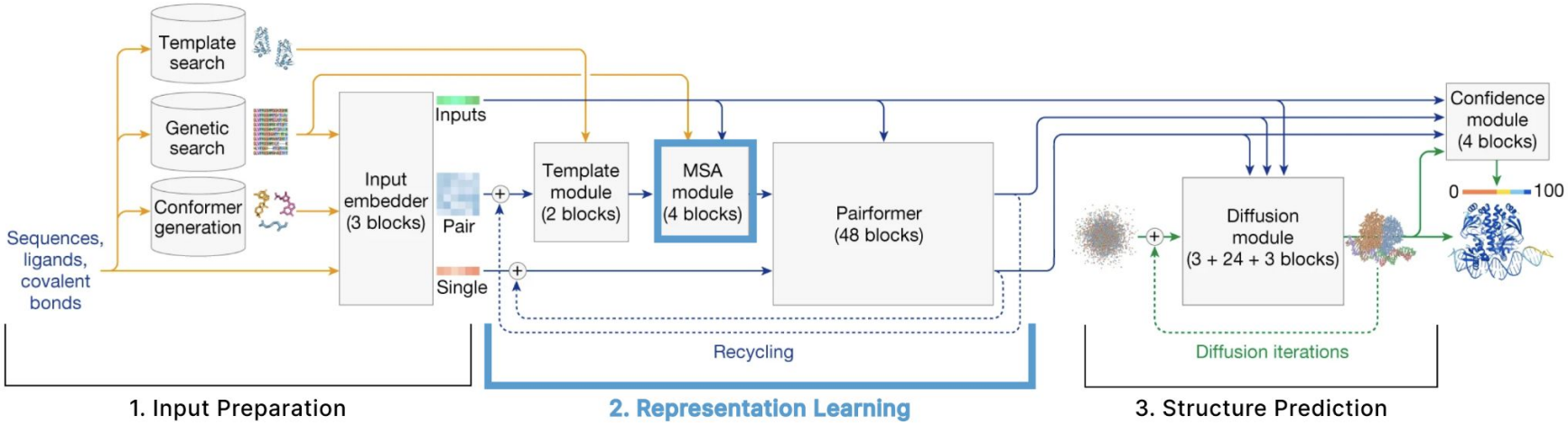
RFDiffusion





https://elanapearl.github.io/blog/2024/the-illustrated-alphafold/

AlphaFold 3



Template Module **MSA module** Pairformer

- **Outer product mean**
- **Row-wise gated attention using only pair bias**

AlphaFold 3

Modeling Interactions

1. **Protein-Nucleic Acid** (DNA/RNA)
2. **Protein-Ligands**
3. **Protein-Protein**

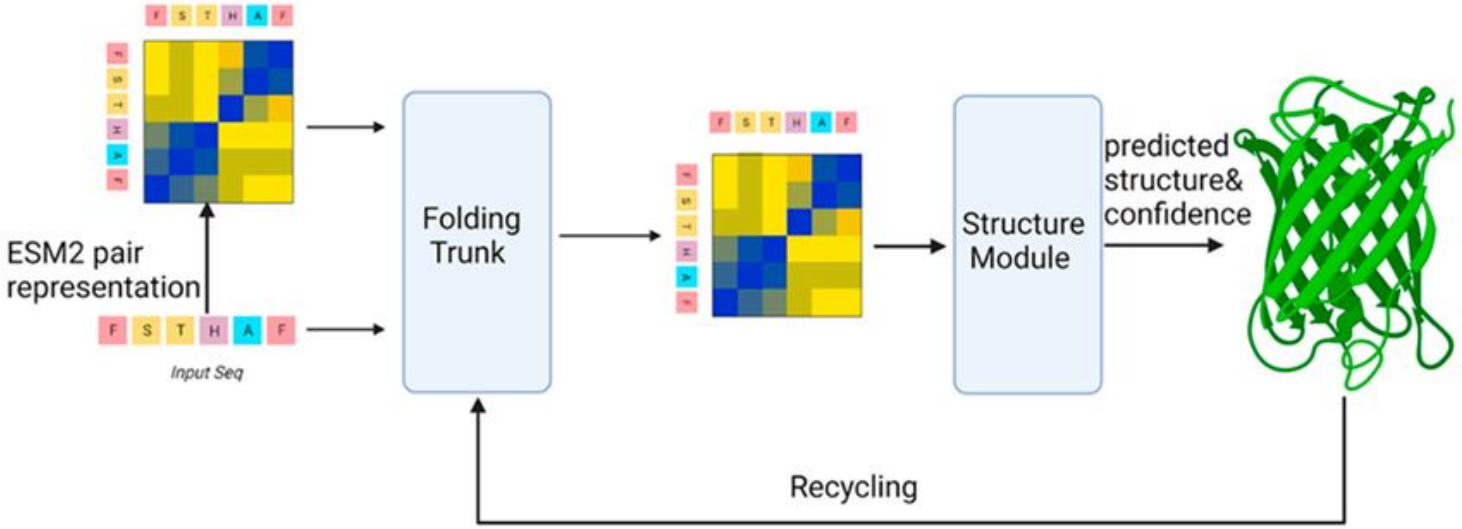
AlphaFold 3

Modeling Interactions

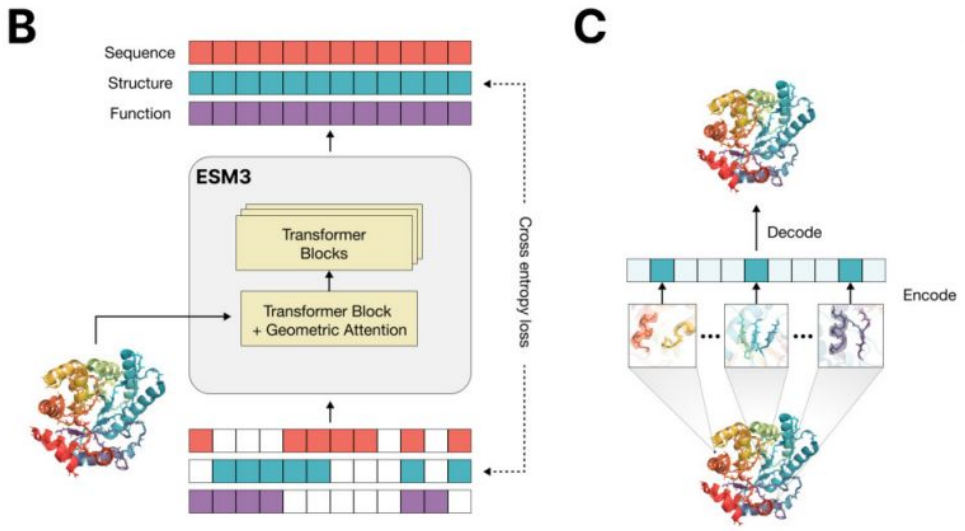
1. **Protein-Nucleic Acid** (DNA/RNA)
2. **Protein-Ligands**
3. **Protein-Protein**

1. **How does proteins interact with DNA; how is it synthesized?**
2. **Genome Editing** (CRISPR-CAS9)
3. Protein interacting with smaller molecules
4. **Antigen-antibody** Understand how the body reacts to pathogens

Evolutionary Scale Fold (ESM 1-2)

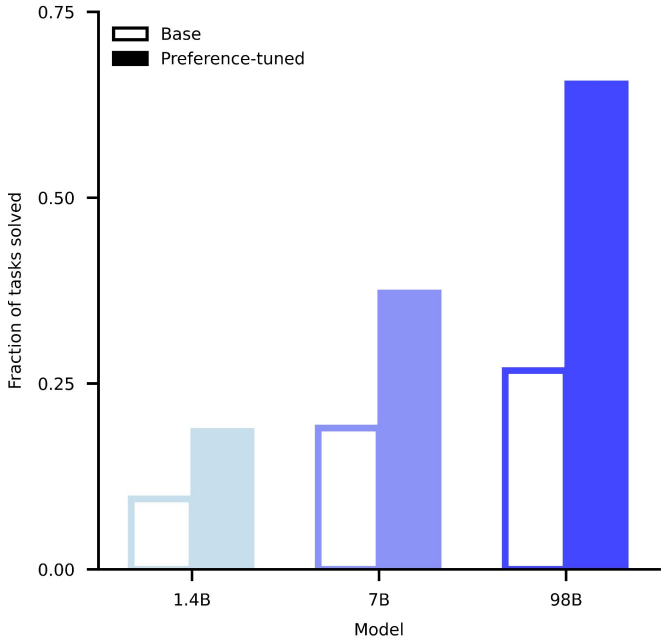


ESM3: Simulating Evolution



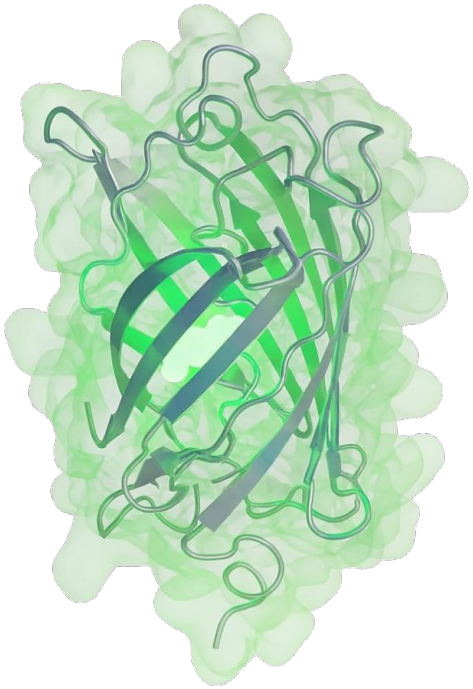
1. Reason over *Sequence, Structure, Function*
2. Simpler LLM
   Simple prompting
3. Biological Fidelity
4. Better generalizing

ESM3: Simulating Evolution



1. Reason over *Sequence, Structure, Function*
2. Simpler LLM
   Simple prompting
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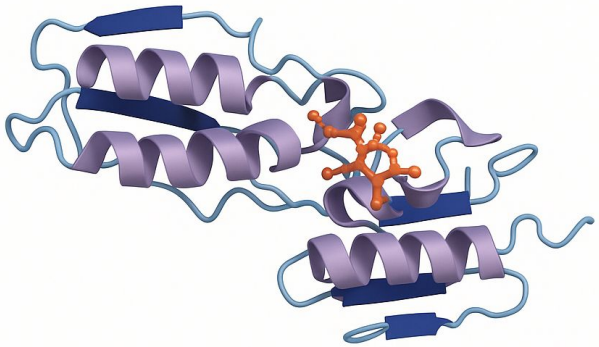
ESM3: New Fluorescent Protein



1. In vitro creation of new Green Fluorescent Proteins (jellyfish)
2. Use CoT to generate candidate proteins

3. Similarity of only 58%
4. 500 million years of evolution removed from the closest known protein

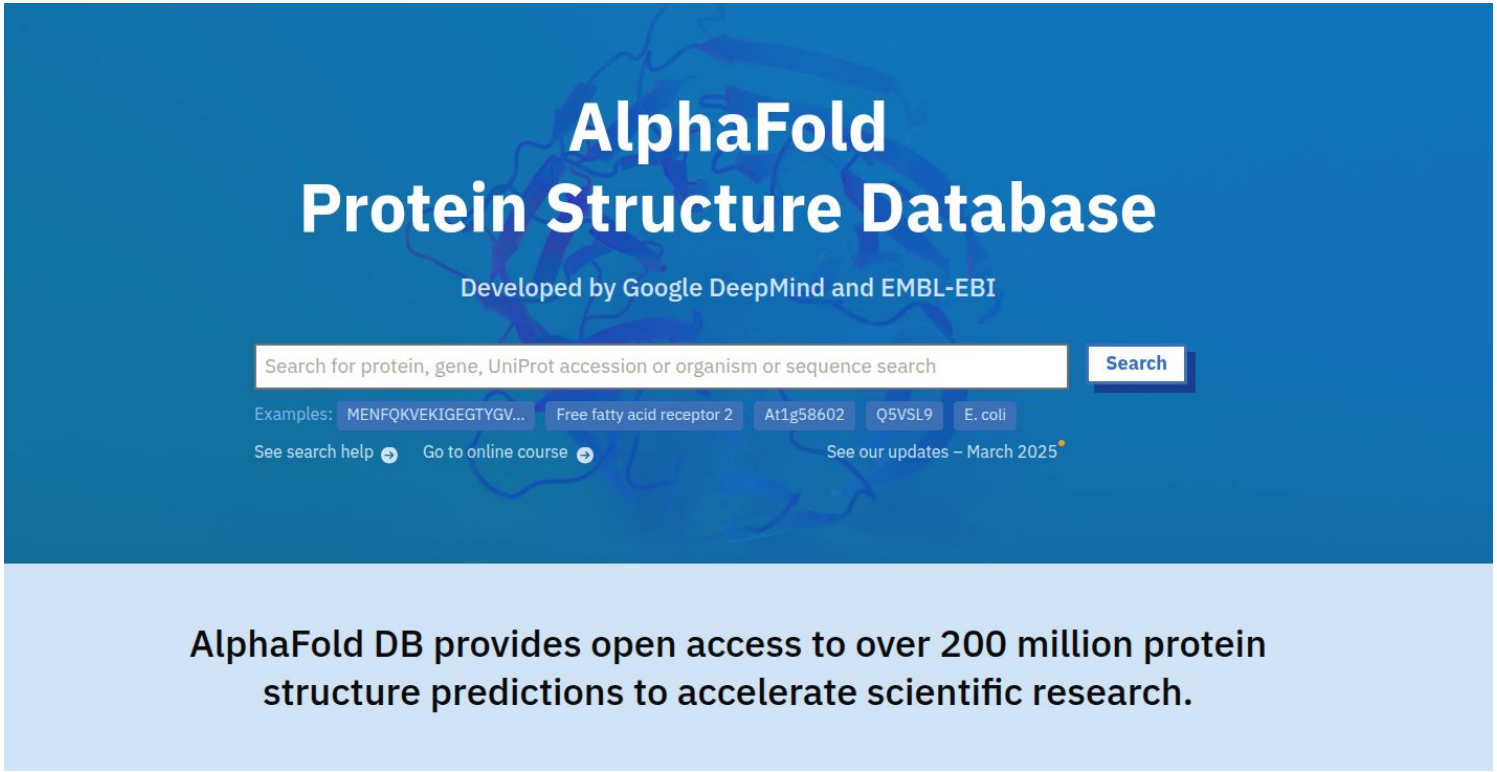
ESM3: New Fluorescent Protein



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4. 500 million years of evolution removed from the closest known protein

AlphaFold Protein Data Bank



AlphaMissense

A catalogue of genetic mutations to help pinpoint the cause of diseases

19 SEPTEMBER 2023

Žiga Avsec and Jun Cheng

Share

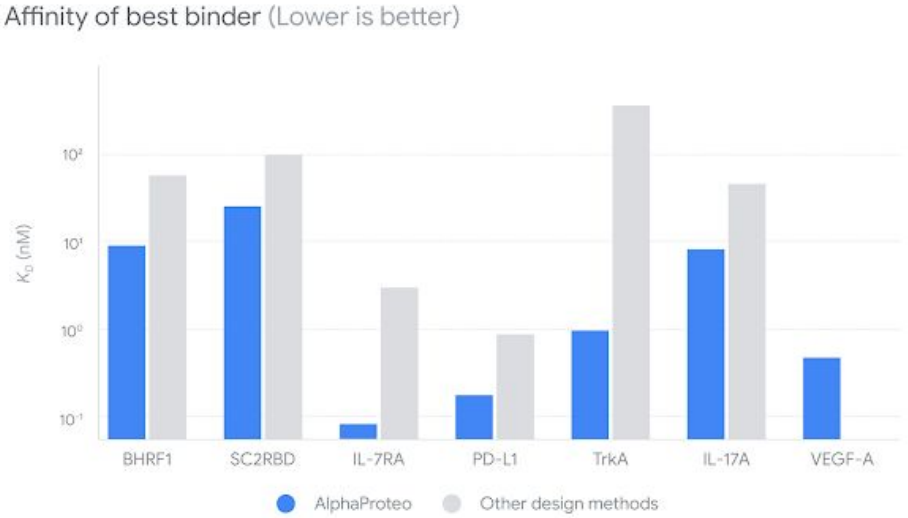


Missense: Mutation in DNA that leads to slightly different protein synthesis

Single letter substitution

Classify 71 million *missense* mutations: Pathogenicity

AlphaProteo: Drug Design and Disease Understanding



AlphaProteo: Drug Design and Disease Understanding

CASE STUDY
Solving one of biology's greatest puzzles
Learn more



CASE STUDY
Breaking down plastic pollution
Learn more



CASE STUDY
Designing more effective drugs
Learn more



CASE STUDY
Increasing honeybees' chances of survival
Learn more



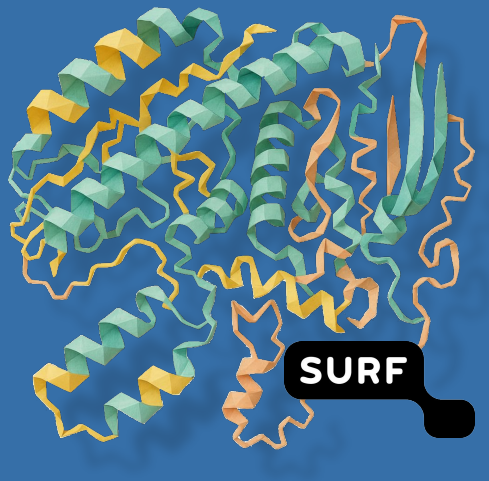
CASE STUDY
Searching for life-changing medicines
Learn more



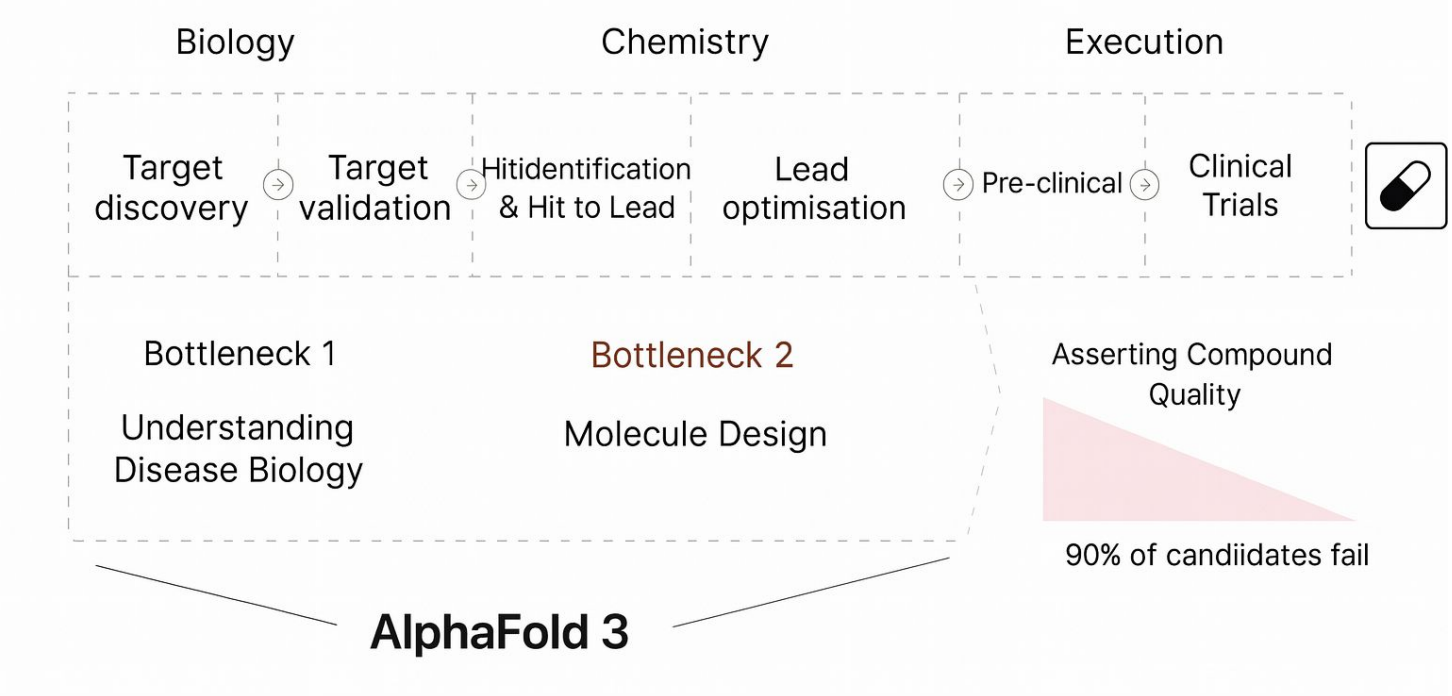
CASE STUDY
Tracing proteins back to origins of life
Learn more



Shape of things to come

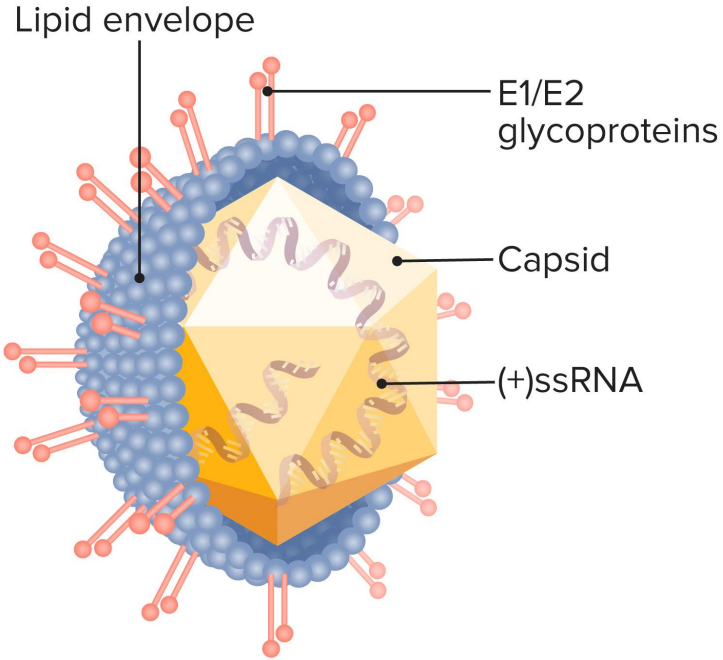


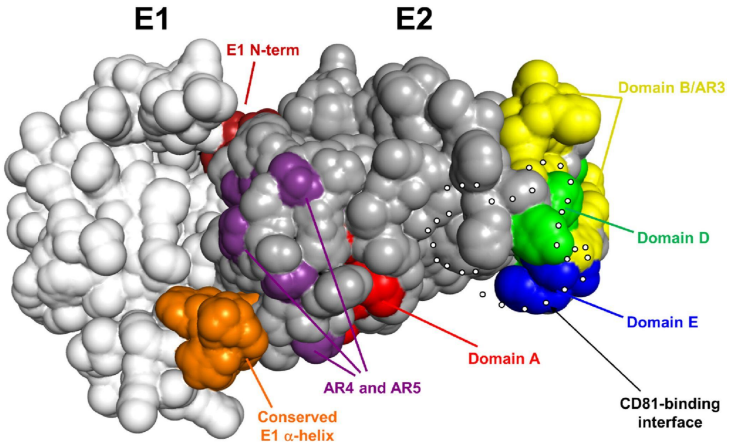
AlphaProteo: Drug Design and Disease Understanding



Experimental vaccinology@AMC

Hepatitis C Virus

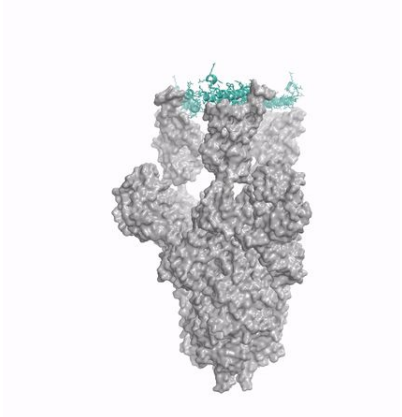




Chaining Predictions

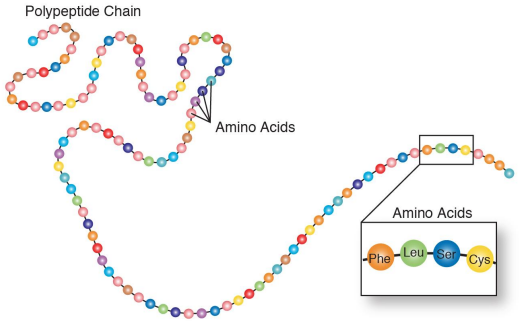
01. Generation

RFDiffusion



02. Predict Sequence

ProteinMPNN



03. Confirm Structure

ESMFold, ColabFold, AlphaFold, OmegaFold, RoseTTaFold, OpenFold



SURF

Thanks!